

Medial-axis reconstruction in hyperbolic space, with a structural framework for Hadamard manifolds*

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Abstract

We generalise Theorem 6 of Białożył [1] — the characterisation of the points reconstructible from the medial axis of a closed set — from Euclidean space to real hyperbolic space $\mathbb{H}^n$, and set up the structural framework for general Hadamard manifolds. For a graph-type closed set $X \subseteq \mathbb{H}^n$ we prove (Theorem 1.1)

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{ \mathcal{H} \text{ maximal free, horospherically defective} \},$$

where a maximal free horoball is *horospherically defective* when its contact set is not convex inside the intrinsically flat horosphere $\cong \mathbb{E}^{n-1}$. The proof rests on two pillars. A curvature-free *reconstruction engine* — a rigidity statement, a drift theorem through the horofunction boundary, and an unconditional structure trichotomy — reduces reconstruction outside the hull to drift into a maximal free horoball. A *horospherical dictionary* special to $\mathbb{H}^n$, whose horospheres are flat, then identifies the feet of high centres with Euclidean nearest contacts, so that Motzkin’s theorem (a closed Euclidean set is convex iff Chebyshev) converts reconstruction into non-convexity of the contact set. Sufficiency follows because a non-convex contact set furnishes medial centres reconstructing the whole horoball; necessity, for compact X , from a deepest-direction shadow lemma, and for non-compact X from an ascent-and-no-collapse analysis of a *moving* contact diagram whose asymptotic contacts are read at finite height. Part II records the genesis of the statement — the naive hull-based notion of [2] fails already for a unit circle in $\mathbb{H}^3$ — together with the worked examples and the obstructions to leaving $\mathbb{H}^n$.

1 Introduction

Białożył’s Theorem 6 [1] describes, for a closed set $X \subseteq \mathbb{R}^n$, exactly which points can be recovered as centres of maximal balls touching X — the *reconstructible set* $\mathcal{R}_X$ — in terms of the convex hull of X and the *defective* supporting half-spaces, those whose contact slice is non-convex. The companion note [2] lifted the sufficiency half to Hadamard manifolds, replacing half-spaces by maximal free horoballs and the convex hull by the horoball hull $\widehat{X}$. Its Conjecture 5.1 proposed that, conversely, every reconstructible point outside $\widehat{X}$ lies in a *defective* horoball, defectiveness being measured against $\widehat{X}$. This note settles the necessity direction in $\mathbb{H}^n$, and in doing so corrects the notion of defect.

*Companion note to [2], which generalised the sufficiency half of Theorem 6 of [1] to Riemannian manifolds, and to our formalization notes [3] on the Euclidean case. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand. Remaining errors are ours.

The main theorem. Throughout, a closed set is *graph-type* if, near every part of it that approaches a horosphere, it is the graph of a locally semiconvex height deficit (the zero set of the deficit being the contacts). All sets arising in practice — smooth or piecewise-smooth hypersurfaces, finite point sets, and their unions — are graph-type.

Theorem 1.1 (Reconstruction in $\mathbb{H}^n$). *Let $X \subseteq \mathbb{H}^n$ be closed and graph-type. Then*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\mathcal{H} \text{ maximal free, horospherically defective}\},$$

where a maximal free horoball $\mathcal{H}$ is horospherically defective if its contact set is not convex inside the flat horosphere $S \cong \mathbb{E}^{n-1}$ — the asymptotic contacts being read through the finite-height moving contact set of §6.2. For compact X this is Corollary 7.2; the inclusion $\supseteq$ is §5 and $\subseteq$ is §6.

Architecture of the proof. Two layers do the work, and separating them is the organising idea of the note.

The reconstruction engine (§3) is purely metric and curvature-free. Its centrepiece is a rigidity statement (Proposition 3.3): one cannot inflate a medial ball along a foot-geodesic, because a second foot cuts the inflation instantly. So multiple contacts on a limit horosphere are never produced by a single inflation — they are produced by *drift*, medial balls of unbounded radius whose centres run off to the ideal boundary. A drift theorem (Theorem 3.6), proved through the horofunction boundary, shows such a drift converges to a maximal free horoball carrying the limiting feet as contacts; and an unconditional structure trichotomy (Theorem 3.8) shows every point outside $\widehat{X}$ either drifts or escapes along a single-contact ray. This reduces the whole question to the geometry of a maximal free horoball and its contacts.

The horospherical dictionary (§4) is special to $\mathbb{H}^n$, whose horospheres are intrinsically flat. Normalising a maximal free horoball to $\{z > 1\}$ in the upper half-space model, the distance from a high centre (α, t) to a contact is monotone in the *Euclidean* distance from α to the contact's horospherical position. Feet are therefore Euclidean nearest contacts, medial centres are non-Chebyshev points, and Motzkin's theorem — a closed Euclidean set is convex iff every point has a unique nearest — turns the entire problem into the convexity of the contact set. Sufficiency (§5) and necessity (§6) are then two faces of this duality.

What is new, and the correction. The hull-based defect of [2] is inadequate: the horoball hull can collapse to X (already for two points in $\mathbb{H}^2$), and in $\mathbb{H}^3$ a unit circle on a horosphere reconstructs an entire region while every maximal free horoball meets it in a connected — hence non-split, non-defective — set (§11). The operative condition is not a gap in the contact set but its failure to be convex inside the flat horosphere; in dimension two these coincide, which is why the phenomenon is invisible there. For non-compact X even the convex hull of the attained contacts is not enough: a single asymptotic ridge and two parallel ones share every directional invariant yet reconstruct oppositely (§12.3). The right reading is through a *moving* contact diagram, in which an asymptotic ray presents at height t as a contact at a moving Euclidean site; the diagram is non-convex exactly when reconstruction occurs.

Organisation. Part I proves Theorem 1.1: the engine (§3), the dictionary (§4), sufficiency (§5), necessity (§6), and the assembly (§7). Part II is the record behind the statement: the genesis through bitangency, splitness and the dimension-three counterexample (§8); the further

results and development notes, including the earlier necessity routes and the boundary-envelope analysis (§12); the move toward general Hadamard manifolds (§13); and a status table (§14). All numerical claims were checked directly; all arguments were re-derived by hand.

We use one external input, as in [2].

Lemma 1.2 (Cut-locus density; Wolter, Albano [6, 5]). *Let M be complete and X closed. The cut locus $\text{Cut}(X)$ — the set of endpoints of maximal minimizing segments of normal geodesics from X — satisfies $\text{Cut}(X) \subseteq \overline{M_X}$.*

Part I. The reconstruction theorem in $\mathbb{H}^n$

2 Setting and definitions

Throughout, (M^n, g) is a *Hadamard manifold*: complete, simply connected, of nonpositive sectional curvature. We write $d(y, z)$ for the Riemannian distance. Every geodesic is globally minimizing, extends to a complete geodesic line, and is determined by a point and a velocity. $X \subsetneq M$ is a fixed nonempty closed set, $d(y) := \text{dist}(y, X)$, and

$$m(y) := \{x \in X \mid d(y, x) = d(y)\}$$

is the nonempty compact set of *feet* of y (Hopf–Rinow). All balls $B(y, r)$ are open. A ball (or any set) is *free* if it is disjoint from X ; thus $B(y, r)$ is free iff $r \leq d(y)$.

Definition 2.1 (Medial axis, reconstructible set). *The medial axis is $M_X := \{a \in X^c : \#m(a) \geq 2\}$; on a Hadamard manifold this coincides with the set of points admitting at least two distinct minimizing geodesics to X (no branching, cf. [2, Rem. 2.2]). The reconstructible set is*

$$\mathcal{R}_X := \bigcup_{a \in M_X} B(a, d(a)).$$

For $p \in \mathcal{R}_X$ and $a \in M_X$ with $p \in B(a, d(a))$ we call $\mu(a, p) := d(a) - d(p, a) > 0$ the *margin* of p in the medial ball $B(a, d(a))$.

Definition 2.2 (Busemann function, maximal free horoball). *For a unit-speed geodesic ray $\gamma : [0, \infty) \rightarrow M$ the Busemann function is $b_\gamma(y) := \lim_{t \rightarrow \infty} (d(y, \gamma(t)) - t)$; the limit exists since the expression is non-increasing in t and bounded below by $-d(y, \gamma(0))$. It is 1-Lipschitz, convex, and $b_\gamma(\gamma(s)) = -s$. Set $c_\gamma := \inf_X b_\gamma$. When c_γ is finite, the maximal free horoball and its horosphere are*

$$\mathcal{H}_\gamma := \{b_\gamma < c_\gamma\}, \quad S_\gamma := \{b_\gamma = c_\gamma\}.$$

$\mathcal{H}_\gamma$ is open, convex, free, and maximal among free sub-level sets of b_γ . The *depth* of a point $p \in \mathcal{H}_\gamma$ is $c_\gamma - b_\gamma(p) > 0$. The *horoball hull* is $\widehat{X} := M \setminus \bigcup \{\text{free open horoballs}\} = \{y : b_\gamma(y) \geq c_\gamma \text{ for every ray } \gamma\}$, a closed set containing X .

The correct notion of defect — forced by the dimension-three counterexample of §11 and motivated there — is intrinsic to the horosphere. Since a horosphere of a Hadamard manifold is intrinsically flat, $S_\gamma \cong \mathbb{E}^{n-1}$, its contact set has an unambiguous flat convex hull.

Definition 2.3 (Horospherical defect). *For a maximal free horoball $\mathcal{H}_\gamma$ let $K_\gamma := S_\gamma \cap X$ be its contact set and let $\text{hconv}_\gamma(K_\gamma)$ be the closed convex hull of K_γ taken in the intrinsic flat metric of the horosphere $S_\gamma \cong \mathbb{E}^{n-1}$. Call $\mathcal{H}_\gamma$ *horospherically defective* if $\text{hconv}_\gamma(K_\gamma) \neq K_\gamma$.*

3 The reconstruction engine

The first layer is metric and curvature-free: it holds on any Hadamard manifold and reduces reconstruction outside the hull to the geometry of a single maximal free horoball. We record the inflation construction of [2, Thm. 3.1] in the quantitative form we need, then prove the rigidity that channels all multiplicity into drift, the drift theorem, and the structure trichotomy.

We first record the inflation construction of [2, Thm. 3.1] in the quantitative form we need. Fix $a \in X^c$, a foot $q \in m(a)$, and let $\gamma : [0, \infty) \rightarrow M$ be the unit-speed ray with $\gamma(0) = q$ and $\gamma(\rho) = a$, where $\rho := d(a)$ (the geodesic from q through a , extended; on a Hadamard manifold the extension is a globally minimizing ray). Define the set of *free times* $T := \{t > 0 : B(\gamma(t), t) \cap X = \emptyset\}$ and the *escape time*

$$r(a, q) := \sup T \in [\rho, \infty].$$

Lemma 3.1 (Monotone free inflation). *T is an interval containing $(0, \rho]$, and for $t' < t$ with $t \in T$ one has $B(\gamma(t'), t') \subseteq B(\gamma(t), t)$. Moreover for every $t \in T$:*

- (i) $d(\gamma(t)) = t$, with $q \in m(\gamma(t))$;
- (ii) if $p \in B(a, \rho)$ with margin $\mu = \rho - d(p, a)$, then $d(p, \gamma(t)) \leq t - \mu$ for all $t \geq \rho$, $t \in T$;
- (iii) every $q' \in m(a)$ satisfies $d(q', \gamma(t)) = t$; in particular all feet of a lie on $\partial B(\gamma(t), t)$.

Proof. Since $d(\gamma(t'), \gamma(t)) = t - t'$ (geodesics minimize), for $y \in B(\gamma(t'), t')$ we get $d(y, \gamma(t)) \leq d(y, \gamma(t')) + (t - t') < t$, proving the nesting and that T is an interval; $\rho \in T$ because $B(a, \rho)$ is free. (i): freeness gives $d(\gamma(t)) \geq t$, while $d(\gamma(t), q) = t$ gives $\leq$ and exhibits q as a foot. (ii): $d(p, \gamma(t)) \leq d(p, a) + d(a, \gamma(t)) = (\rho - \mu) + (t - \rho)$. (iii): $d(q', \gamma(t)) \leq d(q', a) + (t - \rho) = \rho + (t - \rho) = t$ by the triangle inequality, and $\geq t$ by freeness, since $q' \in X$. $\square$

Part (iii) of Lemma 3.1 looks like an engine for producing horoballs with several contacts; in fact it is self-limiting. Boundary-riding is an *equality* in the triangle inequality, and equalities in uniquely geodesic spaces force collinearity, which destroys freeness at once. This is the rigidity that any necessity mechanism must respect.

Lemma 3.2 (Cut branch). *Suppose $r := r(a, q) < \infty$. Then $c := \gamma(r)$ satisfies $d(c) = r$, the ball $B(c, r)$ is free and contains $B(a, \rho)$, and $c \in \text{Cut}(X) \subseteq \overline{M_X}$. Consequently, if $p \in B(a, \rho)$ has margin μ , then for every $\varepsilon \in (0, \mu/2)$ there exists $a' \in M_X$ with*

$$d(a') \geq r - \varepsilon, \quad p \in B(a', d(a')), \quad \mu(a', p) \geq \mu - 2\varepsilon.$$

Proof. $d(\gamma(t)) = t$ on T and continuity of d give $d(c) = r$. If $y \in B(c, r)$, pick $\eta > 0$ with $d(y, c) < r - \eta$; for $t < r$ with $r - t < \eta/2$, $d(y, \gamma(t)) \leq d(y, c) + (r - t) < r - \eta + \eta/2 < t$, so $y \in B(\gamma(t), t)$, which is free; hence $B(c, r)$ is free, and it contains each $B(\gamma(t), t) \supseteq B(a, \rho)$ by Lemma 3.1. The segment $\gamma|_{[0, t]}$ minimizes $d(\cdot, X)$ for every $t \leq r$ by Lemma 3.1(i), while for $t > r$ freeness fails, so $d(\gamma(t)) < t$ and the segment no longer minimizes: c is the cut point of X along γ , and Lemma 1.2 applies. Finally take $a' \in M_X$ with $d(a', c) < \varepsilon$: then $d(a') \geq d(c) - \varepsilon = r - \varepsilon$ and $d(p, a') \leq d(p, c) + \varepsilon \leq (r - \mu) + \varepsilon$ using Lemma 3.1(ii) at $t \rightarrow r$; hence $\mu(a', p) = d(a') - d(p, a') \geq \mu - 2\varepsilon > 0$. $\square$

Proposition 3.3 (Instant cutting). *Let M be Hadamard and $a \in M_X$. Then $r(a, q) = d(a)$ for every foot $q \in m(a)$: the inflation of a medial ball along any of its foot-geodesics is cut instantly.*

Proof. Write $\rho := d(a)$, pick a second foot $q' \neq q$, and suppose $t \in T$ with $t > \rho$. By Lemma 3.1(iii), $d(q', \gamma(t)) = t = d(q', a) + d(a, \gamma(t))$: equality in the triangle inequality, so the concatenation of the geodesic segments $[q', a]$ and $[a, \gamma(t)]$ has length $d(q', \gamma(t))$ and is therefore the (unique) geodesic from q' to $\gamma(t)$. In particular it passes through a with velocity $\dot{\gamma}(\rho)$, so its initial segment $[q', a]$, reversed, is part of γ : $q' = \gamma(\rho - d(q', a)) = \gamma(0) = q$, contradicting $q' \neq q$. Hence $T \subseteq (0, \rho]$ and $r(a, q) = \rho$. $\square$

What survives of the escape mechanism is its single-contact version, which by Proposition 3.3 can only fire at uniquely-footed centres; it supplies the third branch of the structure theorem.

Theorem 3.4 (Single-foot escape). *Let M be Hadamard, X closed, $a \in X^c$, $q \in m(a)$, and let $p \in B(a, d(a))$ have margin $\mu > 0$. If $r(a, q) = \infty$ — which forces $m(a) = \{q\}$ — then with γ the ray of §3:*

- (a) $c_\gamma = 0$, attained at q : $\mathcal{H}_\gamma = \{b_\gamma < 0\}$ is a maximal free horoball with attained contact $q \in S_\gamma \cap X$;
- (b) $b_\gamma(p) \leq -\mu$, so $p \in \mathcal{H}_\gamma$ at depth at least μ ;
- (c) $\mathcal{H}_\gamma = \bigcup_{t>0} B(\gamma(t), t)$, an increasing union of free balls each containing $B(a, d(a))$ for $t \geq d(a)$.

Proof. All balls $B(\gamma(t), t)$, $t > 0$, are free. For $x \in X$ this means $d(x, \gamma(t)) \geq t$, hence $b_\gamma(x) \geq 0$ and $c_\gamma \geq 0$. By Lemma 3.1(i), $d(q, \gamma(t)) = t$ for all t , so $b_\gamma(q) = 0$; with $q \in X$ this forces $c_\gamma = 0$, attained at q , proving (a). By Lemma 3.1(ii), $d(p, \gamma(t)) \leq t - \mu$ for all $t \geq d(a)$, whence $b_\gamma(p) \leq -\mu < 0 = c_\gamma$, proving (b). For (c), the inclusion $\subseteq$ of the union follows from $b_\gamma(y) \leq d(y, \gamma(t)) - t < 0$ for $y \in B(\gamma(t), t)$; conversely if $b_\gamma(y) < 0$ then $d(y, \gamma(t)) < t$ for all large t . The containment statement is Lemma 3.1. $\square$

By Proposition 3.3, escaping inflations cannot start from medial centres; the mechanism by which the medial axis actually produces multi-contact horoballs is *drift* — medial balls containing p with radii tending to infinity and uniform margin, as in the bisector family of a two-point set in $\mathbb{H}^2$. For this branch we pass to the horofunction description of the boundary at infinity. For $\delta > 0$ set

$$T_\delta(p) := \sup\{d(a) : a \in M_X, \mu(a, p) \geq \delta\} \in [0, \infty], \quad \sup \emptyset := 0.$$

Lemma 3.5 (Horofunction convergence; [4, II.8.13]). *Let M be Hadamard, $o \in M$, and let $(a_k) \subset M$ with $d(o, a_k) \rightarrow \infty$ and $a_k \rightarrow \xi \in \partial_\infty M$ in the cone topology. Then $h_k(\cdot) := d(\cdot, a_k) - d(o, a_k) \rightarrow b_\xi(\cdot)$ uniformly on compact sets, where b_ξ is the Busemann function of the ray from o to ξ , normalized by $b_\xi(o) = 0$. Moreover $\overline{M} = M \cup \partial_\infty M$ is compact, so every unbounded sequence subconverges to some $\xi \in \partial_\infty M$.*

Theorem 3.6 (Limit horoball from unbounded medial radii). *Let M be Hadamard, X closed, $p \in X^c$, and suppose $T_\delta(p) = \infty$ for some $\delta > 0$. Then there exist $\xi \in \partial_\infty M$, $s_\infty \in [\delta, d(p)]$, medial centres $a_k \in M_X$ with $r_k := d(a_k) \rightarrow \infty$ and $\mu(a_k, p) \geq \delta$, such that, with b_ξ normalized at $o = p$:*

- (a) $X \subseteq \{b_\xi \geq s_\infty\}$ and $b_\xi(p) = 0 \leq s_\infty - \delta$; hence $c_\xi \geq s_\infty > 0 = b_\xi(p)$ and p lies in the maximal free horoball $\mathcal{H}_\xi = \{b_\xi < c_\xi\}$ at depth at least δ ;

- (b) the free balls exhaust the sub-horoball: every compact $K \subseteq \{b_\xi < s_\infty\}$ satisfies $K \subseteq B(a_k, r_k)$ for all large k ;
- (c) (contact capture) if feet $q_k \in m(a_k)$ have a bounded subsequence, then along it $q_k \rightarrow q_* \in X$ with $b_\xi(q_*) = s_\infty$; in that case $c_\xi = s_\infty$, the infimum is attained at q_* , and $q_* \in S_\xi \cap X$. If two sequences of feet $q_k \neq q'_k$ of a_k subconverge to distinct limits, $\mathcal{H}_\xi$ is bitangent.

Proof. Choose $a_k \in M_X$ with $\mu(a_k, p) \geq \delta$ and $r_k = d(a_k) \rightarrow \infty$. Set $t_k := d(p, a_k) \leq r_k - \delta$ and $s_k := r_k - t_k \geq \delta$. For any $x_0 \in X$, freeness of $B(a_k, r_k)$ gives $d(x_0, a_k) \geq r_k$, so $d(x_0, p) \geq d(x_0, a_k) - t_k \geq s_k$; hence $s_k \leq d(p)$, and after passing to a subsequence $s_k \rightarrow s_\infty \in [\delta, d(p)]$. Also $t_k \geq r_k - d(p) \rightarrow \infty$, so (a_k) is unbounded and, after a further subsequence, $a_k \rightarrow \xi \in \partial_\infty M$ (Lemma 3.5).

(a) For $x \in X$: $h_k(x) = d(x, a_k) - t_k \geq r_k - t_k = s_k$, and letting $k \rightarrow \infty$, $b_\xi(x) \geq s_\infty$. Trivially $b_\xi(p) = 0$. Hence $c_\xi = \inf_X b_\xi \geq s_\infty \geq \delta$, while $c_\xi \leq b_\xi(x_0) < \infty$; the maximal free horoball $\mathcal{H}_\xi$ exists and contains p at depth $c_\xi - b_\xi(p) \geq \delta$.

(b) If $b_\xi(y) < s_\infty$ then, by locally uniform convergence on a compact neighbourhood of K , $h_k(y) < s_k$ for large k uniformly on K , i.e. $d(y, a_k) < t_k + s_k = r_k$.

(c) Along the bounded subsequence, $h_k(q_k) = d(q_k, a_k) - t_k = r_k - t_k = s_k$, and locally uniform convergence plus $q_k \rightarrow q_*$ gives $b_\xi(q_*) = s_\infty$. Since $q_* \in X$ (closedness) we get $c_\xi \leq s_\infty$, so $c_\xi = s_\infty$ is attained at q_* . Two distinct such limits give two points of $S_\xi \cap X$. $\square$

Remark 3.7. Without the boundedness hypothesis in (c) the feet may escape to infinity and the contact be only asymptotic ($c_\xi = s_\infty$ not attained, or attained at fewer points than the medial multiplicity suggests). This is a genuine phenomenon already in $\mathbb{R}^2$: for $X = \text{graph}(e^x) \cup \{(5, 0)\}$ the medial centres $(\bar{x}_T, -R_T)$ with feet $(5, 0)$ and $(-T, e^{-T})$ have $R_T \rightarrow \infty$, one foot converging to the attained contact $(5, 0)$ and the other escaping along the asymptote; the limit half-plane $\{y < 0\}$ is maximal free with a single attained contact. See §8.

3.1 The structure trichotomy

The two branches above, with the hull, exhaust the points outside X unconditionally.

Theorem 3.8 (Structure trichotomy — no residual case). *Let M be Hadamard, X closed, and $p \in X^c$ (reconstructible or not). Then at least one of the following holds.*

- (I) (**Hull**) $p \in \widehat{X} \setminus X$.
- (II) (**Drift**) $T_\delta(p) = \infty$ for some $\delta > 0$; then p lies, at depth $\geq \delta$, in a maximal free horoball exhausted (below level s_∞) by medial balls, whose attained contacts are the subsequential limits of their feet (Theorem 3.6).
- (III) (**Ray escape**) There is $\delta' > 0$ such that for all sufficiently large t , p lies at depth $\geq \delta'$ in a maximal free horoball with an attained contact, produced by an escaping single-foot inflation (Theorem 3.4).

Proof. Assume (I) fails; since $p \notin X$, this means $p \notin \widehat{X}$. Choose a ray γ with $b_\gamma(p) < c_\gamma =: c$ (finite by this very inequality) and set $\delta^* := c - b_\gamma(p) > 0$. For $x \in X$ the function $s \mapsto d(x, \gamma(s)) - s$ is non-increasing with limit $b_\gamma(x) \geq c$, so $d(x, \gamma(t)) \geq t + c$ for every t : the balls $B(\gamma(t), t + c)$ are free, and $d(\gamma(t)) \geq t + c$. Since $d(p, \gamma(t)) - t \downarrow b_\gamma(p) = c - \delta^*$, there is t_0 with $d(p, \gamma(t)) \leq t + c - \delta^*/2$ for $t \geq t_0$: p lies in the maximal free ball $B(\gamma(t), d(\gamma(t)))$ with

margin $\mu_t \geq \delta^*/2$. For $t \geq t_0$ pick a foot $q_t \in m(\gamma(t))$. If $r(\gamma(t), q_t) < \infty$ for an unbounded set of t , Lemma 3.2 with $\varepsilon = \delta^*/8$ supplies $a'_t \in M_X$ with $d(a'_t) \geq d(\gamma(t)) - \delta^*/8 \geq t + c - \delta^*/8$ and $\mu(a'_t, p) \geq \delta^*/4$, so $T_{\delta^*/4}(p) = \infty$: case (II). Otherwise $r(\gamma(t), q_t) = \infty$ for all large t , and Theorem 3.4 (applied to $a = \gamma(t)$, margin $\delta^*/2$) gives case (III) with $\delta' = \delta^*/2$. $\square$

Remark 3.9. The trichotomy is unconditional and does not require $p \in \mathcal{R}_X$: reconstructibility enters only when the horoballs delivered by drift or ray-escape are upgraded to defective ones, which is the content of the horospherical dictionary below. It removes the “confinement” case conjectured in an earlier draft.

In Euclidean space the drift branch always carries reconstruction outside the hull, by lifting the defect witness of [3].

Proposition 3.10 (In $\mathbb{R}^n$, reconstruction outside the hull forces drift). *For $M = \mathbb{R}^n$ and $p \in \mathcal{R}_X \setminus \overline{\text{conv}} X$, case (II) of Theorem 3.8 holds: $T_\delta(p) = \infty$ for some $\delta > 0$.*

Proof. By the repaired necessity direction of Theorem 6 [3, §4.6] there are a unit vector v , $c = \sup_X \langle v, \cdot \rangle$, and a defect witness $w \in (\overline{\text{conv}} X \cap \{\langle v, \cdot \rangle = c\}) \setminus X$ with $h := \langle v, p \rangle - c > 0$. Lift the witness: $w_t := w + tv$. Then $d(w_t) \geq \langle v, w_t \rangle - c = t$ since $X \subseteq \{\langle v, \cdot \rangle \leq c\}$, while $|p - w_t|^2 = t^2 - 2th + |p - w|^2$ gives $|p - w_t| \leq t - h + O(1/t)$: for all large t , p lies in $B(w_t, d(w_t))$ with margin $\mu_t \geq h/2$.

No inflation from w_t escapes. Indeed, suppose $r(w_t, q_t) = \infty$ for a foot $q_t \in m(w_t)$, and let u_t be the direction of the inflation ray, so $q_t = w_t - \rho_t u_t$ with $\rho_t := d(w_t)$. Freeness of all balls $B(\gamma_t(s), s)$ gives, for every $x \in X$, $\langle x - q_t, u_t \rangle \leq 0$; the same then holds for every point of $\overline{\text{conv}} X$, in particular for w . Substituting $w - q_t = \rho_t u_t - tv$:

$$0 \geq \langle w - q_t, u_t \rangle = \rho_t - t \langle v, u_t \rangle \geq \rho_t - t \geq 0,$$

forcing equality throughout: $\langle v, u_t \rangle = 1$, hence $u_t = v$, $\rho_t = t$, and $q_t = w_t - tv = w \notin X$ — contradicting $q_t \in X$.

Hence $r(w_t, q_t) < \infty$ for every t large, and Lemma 3.2 with $\varepsilon = h/8$ supplies $a'_t \in M_X$ with $d(a'_t) \geq d(w_t) - h/8 \geq t - h/8$ and $\mu(a'_t, p) \geq h/4$. Thus $T_{h/4}(p) = \infty$. $\square$

Corollary 3.11 (Strengthened Euclidean necessity). *For closed $X \subseteq \mathbb{R}^n$, every $p \in \mathcal{R}_X \setminus \overline{\text{conv}} X$ lies at positive depth in a maximal free half-space L^+ that is a limit of medial balls in the sense of Theorem 3.6(b); the attained contacts of ∂L^+ are the subsequential limits of their feet. In particular the union in Theorem 6 of [1] may be taken over maximal free half-spaces arising as limits of medial balls.*

4 The horospherical dictionary in $\mathbb{H}^n$

The second layer is special to real hyperbolic space, whose horospheres are intrinsically flat. It converts the geometry of a maximal free horoball into Euclidean convexity through a classical theorem.

Theorem 4.1 (Motzkin [7]). *A non-empty closed set $C \subseteq \mathbb{E}^N$ is convex if and only if it is Chebyshev: every point of $\mathbb{E}^N$ has a unique nearest point in C . Equivalently, C is non-convex iff some point has at least two nearest points in C .*

We use the upper half-space model $\mathbb{H}^n = \{(\xi, z) : \xi \in \mathbb{R}^{n-1}, z > 0\}$ with $\cosh d(p, q) = 1 + |p - q|^2 / (2z_p z_q)$. Isometries act transitively on horoballs and restrict to Euclidean similarities of the flat horosphere (preserving convexity), so for any maximal free horoball it costs nothing to assume

$$\mathcal{H} = \{z > 1\}, \quad S = \{z = 1\} \cong \mathbb{R}^{n-1}, \quad X \subseteq \{z \leq 1\}, \quad K = X \cap \{z = 1\}.$$

For $a = (\alpha, t)$ and a contact $k = (\kappa, 1)$, $\cosh d(a, k) = 1 + \frac{|\alpha - \kappa|^2 + (t-1)^2}{2t}$ is strictly increasing in the *Euclidean* position distance $|\alpha - \kappa|$; so among contacts the feet of a are the Euclidean-nearest elements of K , and the medial directions are the non-Chebyshev points of K .

Two structural facts about feet, used in both directions, hold once we are in this picture. First, the feet of a drift ascend to the horosphere; second, over a graph contact they never collapse to a single point. Both follow from the leading-order shape of $\cosh d$ at a high centre.

Lemma 4.2 (Ascent of all feet). *Let $\mathcal{H} = \{z > 1\}$ be the maximal free horoball in direction ∞ , so $\sup_X z = 1$. Let $a_k = (\alpha_k, t_k)$ satisfy $t_k \rightarrow \infty$ and $|\alpha_k| = o(t_k)$ (i.e. $a_k \rightarrow \infty$ in the cone topology — automatic for a drift toward ∞). Then every foot $q_k = (\beta_k, s_k) \in m(a_k)$ has $s_k \rightarrow 1$, whether β_k stays bounded or escapes.*

Proof. Fix $\varepsilon \in (0, 1)$ and, using $\sup_X z = 1$, a point $y_\varepsilon = (\nu_\varepsilon, w_\varepsilon) \in X$ with $w_\varepsilon > 1 - \varepsilon$ (position ν_ε fixed). As q_k is nearest, $\cosh d(a_k, q_k) \leq \cosh d(a_k, y_\varepsilon)$; dropping the non-negative position term on the left and clearing $2t_k$,

$$\frac{(t_k - s_k)^2}{s_k} \leq \frac{|\alpha_k - \nu_\varepsilon|^2 + (t_k - w_\varepsilon)^2}{w_\varepsilon}.$$

Here $|\alpha_k - \nu_\varepsilon|^2 = o(t_k^2)$ (as $|\alpha_k| = o(t_k)$) and $(t_k - s_k)^2, (t_k - w_\varepsilon)^2 = t_k^2(1 + o(1))$; dividing by t_k^2 gives $\frac{1+o(1)}{s_k} \leq \frac{1+o(1)}{w_\varepsilon}$, so $\liminf_k s_k \geq w_\varepsilon > 1 - \varepsilon$. As ε was arbitrary and $s_k \leq 1$, $s_k \rightarrow 1$. $\square$

The companion fact — that two feet over a graph contact stay apart — is the engine of necessity and removes the tangential-bulk obstruction in sufficiency.

Theorem 4.3 (No collapse at a graph contact). *Suppose near every contact the set X is a graph $\{(\chi, 1 - \phi(\chi))\}$ with $\phi \geq 0$ locally semiconvex and $\{\phi = 0\}$ the contact positions. Then no high centre has two feet collapsing to a single point: for $a = (\alpha, t)$ the feet near a contact are isolated minimisers of $F_t(\chi) = |\alpha - \chi|^2 + t^2\phi(\chi)$, and as $t \rightarrow \infty$ each converges to a distinct zero of ϕ , i.e. a distinct contact. In particular the collapse alternative of Theorem 12.4 is vacuous for graph-type X .*

Proof. With $\cosh d(a, (\chi, 1 - \phi)) = 1 + \frac{|\alpha - \chi|^2 + (t-1+\phi)^2}{2t(1-\phi)}$, minimising over χ is, to leading order for large t , minimising $F_t(\chi) = |\alpha - \chi|^2 + t^2\phi(\chi)$ (§4.3 expansion). If ϕ is convex on a neighbourhood, F_t is the sum of the strictly convex $|\alpha - \chi|^2$ and the convex $t^2\phi$, hence strictly convex with a unique minimiser — one foot, even when $\text{Hess } \phi(\kappa_*) = 0$ degenerates (the quartic $\phi = |\chi|^4$ included). If ϕ is not convex, its zero set — the contacts — consists of separated points (a touching set with a gap), and the minimisers of F_t localise, as $t \rightarrow \infty$, to these separated zeros: two feet sit over two *distinct* contacts at a fixed separation. Either way feet never coalesce to one point. $\square$

5 Sufficiency: a defective horoball reconstructs its interior

We prove the inclusion $\supseteq$ of Theorem 1.1: a horospherically defective maximal free horoball lies in $\mathcal{R}_X$. The mechanism is uniform — a non-Chebyshev point of the contact set is a medial direction at every height, and high enough centres sweep the whole horoball. We give it first when X lies on the horosphere (where it is sharp), then when the contacts are isolated from the bulk, then for compact X in general.

Theorem 5.1 (Reconstruction on a horosphere). *Let X lie on a horosphere of $\mathbb{H}^n$ (so $K = X$). Then:*

- (a) *if X is not convex in $S \cong \mathbb{E}^{n-1}$, the entire horoball is reconstructed, $\mathcal{H} \subseteq \mathcal{R}_X$;*
- (b) *if X is convex, then $M_X = \emptyset$ and $\mathcal{R}_X = \emptyset$.*

Proof. Every point of X has height 1, so for any $a = (\alpha, t)$ the feet $m(a)$ are exactly the Euclidean-nearest positions of $K = X$ to α , for every $t > 0$.

(b) If X is convex, Motzkin gives a unique nearest position for each α , so every a has a single foot: $M_X = \emptyset$.

(a) If X is non-convex, Motzkin gives α^* with two distinct nearest contacts k_1, k_2 at position distance $\rho := \rho_K(\alpha^*) > 0$. Then $a_t := (\alpha^*, t)$ has $k_1, k_2 \in m(a_t)$ for every $t > 0$: it is medial, with $\cosh d(a_t) = 1 + \frac{\rho^2 + (t-1)^2}{2t}$. For $p = (\pi, z) \in \mathcal{H}$ (so $z > 1$), $p \in B(a_t, d(a_t))$ iff $\cosh d(p, a_t) < \cosh d(a_t)$, which after clearing denominators and cancelling $-2zt$ reads

$$(1 - z)t^2 + (|\pi - \alpha^*|^2 + z^2 - z\rho^2 - z) < 0.$$

Since $z > 1$ the leading coefficient $1 - z$ is negative, so the inequality holds for all t large enough: $p \in \mathcal{R}_X$. As $p \in \mathcal{H}$ was arbitrary, $\mathcal{H} \subseteq \mathcal{R}_X$. $\square$

Theorem 5.2 (Reconstruction, isolated contacts). *Let $\mathcal{H} = \{z > 1\}$ be maximal free in $\mathbb{H}^n$ with contact set K , and suppose the contacts are isolated from the bulk: $X \cap \{z > 1 - \nu\} = K$ for some $\nu > 0$. If K is non-convex in $S \cong \mathbb{E}^{n-1}$ (i.e. $\mathcal{H}$ is horospherically defective), then $\mathcal{H} \subseteq \mathcal{R}_X$.*

Proof. Take α^* and k_1, k_2, ρ as before (Motzkin on the closed set K). For $x = (\chi, s) \in X$ with $s \leq 1 - \nu$,

$$\cosh d(a_t, x) - \cosh d(a_t, k_1) \geq \frac{(t-s)^2}{2ts} - \frac{\rho^2 + (t-1)^2}{2t} = \frac{(1-s)(t^2 - s) - s\rho^2}{2ts},$$

and $(1-s)(t^2 - s) - s\rho^2 \geq \nu(t^2 - 1) - \rho^2 > 0$ once $t > \sqrt{1 + \rho^2/\nu}$ — a bound independent of x . So beyond this t no bulk point beats the nearest contacts, $m(a_t) \subseteq K$ consists of the (two or more) Euclidean-nearest contacts, and a_t is medial. The reconstruction inequality of Theorem 5.1(a) is unchanged (it used only the contacts, at height 1), so $\mathcal{H} \subseteq \mathcal{R}_X$. $\square$

Theorem 5.3 (Sufficiency for compact sets). *Let $X \subseteq \mathbb{H}^n$ be compact and graph-type at its contacts. Then every horospherically defective maximal free horoball $\mathcal{H}$ satisfies $\mathcal{H} \subseteq \mathcal{R}_X$.*

Proof. Normalise $\mathcal{H} = \{z > 1\}$; K is non-convex, so by Motzkin it has a non-Chebyshev point. The feet of a high centre $a = (\alpha, t)$ are, by Theorem 4.3, one isolated minimiser of F_t per local contact cluster — the tangential bulk that shadowed a *fixed* direction in Remark 12.8(i)

merely shifts each minimiser slightly, without creating or destroying one. Hence the nearest-foot map is single-valued off a separating set, and as α crosses from the basin of one contact cluster to another the two competing nearest distances exchange order; by continuity there is α^* with two equidistant nearest feet — a medial high centre. Running this for $t \rightarrow \infty$, the reconstruction inequality of Theorem 5.1(a) (which used only that the two feet sit at height $\rightarrow 1$) gives $p \in B(a, d(a))$ for every $p \in \mathcal{H}$ once t is large. So $\mathcal{H} \subseteq \mathcal{R}_X$. $\square$

6 Necessity: a reconstructible point sits in a defective horoball

We prove the inclusion $\subseteq$ of Theorem 1.1. For compact X a single first-order argument at the deepest direction does it, with no drift analysis at all. For non-compact X the contacts can recede to the ideal boundary, and the right bookkeeping is a moving Euclidean diagram.

6.1 The compact case: the deepest-shadow lemma

Let $X \subseteq \mathbb{H}^n$ be compact, $p \notin \widehat{X}$, and $D(\xi) := c_\xi - b_\xi(p)$ the depth of p in the maximal free horoball at the ideal point ξ . As D is upper semicontinuous on the compact $\partial_\infty \mathbb{H}^n$ and positive somewhere, it attains its maximum $\delta^* = D(\xi^*) > 0$. Normalise $\xi^* = \infty$, so $\mathcal{H}^* = \{z > 1\}$, $p = (\pi, z_p)$ with $z_p = e^{\delta^*}$, and $K^* = X \cap \{z = 1\}$. The deepest horoball is already defective.

Lemma 6.1 (Deepest shadow). $\pi \in \text{conv}(K^*)$ in $\mathbb{R}^{n-1}$.

Proof. Suppose not: a unit $u \in \mathbb{R}^{n-1}$ separates, $\langle u, \kappa \rangle \leq \langle u, \pi \rangle - \varepsilon$ for all $\kappa \in K^*$ and some $\varepsilon > 0$. Tilt the direction from ∞ toward the finite ideal point Ru and let $R \rightarrow \infty$. A normalised computation of the Busemann function gives, uniformly on compacta,

$$b_{Ru}(y) = b_\infty(y) - \frac{2}{R} \langle y', u \rangle + O(R^{-2}), \quad y = (y', y_z), \quad b_\infty = -\log y_z.$$

Hence $D(Ru) = \inf_{x \in X} [(b_\infty(x) - b_\infty(p)) - \frac{2}{R} \langle x' - \pi, u \rangle] + O(R^{-2})$. Split X at a small height threshold $\rho > 0$. On contacts and near-contacts ($b_\infty(x) < \rho$, height $> e^{-\rho}$): by compactness their positions cluster on K^* as $\rho \rightarrow 0$, so $\langle x' - \pi, u \rangle \leq -\varepsilon/2$, and the bracket is $\geq \delta^* + \varepsilon/R$. On the bulk ($b_\infty(x) \geq \rho$, positions bounded by $\text{diam } X =: C$): the bracket is $\geq \delta^* + \rho - 2C/R \geq \delta^* + \rho/2$ for large R . Thus $D(Ru) \geq \delta^* + \min(\varepsilon/R, \rho/2) > \delta^*$, contradicting the maximality of ξ^* . $\square$

Lemma 6.2 (Screening). *If $\pi \in K^*$, then $(\pi, 1)$ is the unique nearest point of X to p , and $p \notin \mathcal{R}_X$.*

Proof. For $x = (\chi, s) \in X$, $s \leq 1$, $\cosh d(p, x) - 1 = \frac{|\pi - \chi|^2 + (z_p - s)^2}{2z_p s} \geq \frac{(z_p - s)^2}{2z_p s}$, and $g(s) = (z_p - s)^2/s$ is strictly decreasing on $(0, z_p)$, so for $s < 1$ it exceeds $g(1) = (z_p - 1)^2$; for $s = 1$ it equals $|\pi - \chi|^2 + (z_p - 1)^2$, minimised only at $\chi = \pi$. So $(\pi, 1) \in K^* \subseteq X$ is the unique nearest point. Then d is differentiable at p with $\nabla d(p)$ the upward vertical, and the entire vertical ray $\sigma(\tau) = (\pi, z_p e^\tau)$ keeps $(\pi, 1)$ as unique nearest, with $d(\sigma(\tau)) = \delta^* + \tau \rightarrow \infty$: a steepest-ascent trajectory of d with a single foot, escaping to infinity. By the gradient-flow characterisation of medial-axis reconstruction [8] (valid for distance functions to closed sets in Hadamard manifolds via the semiconcavity of d [9]), such a point lies in no medial ball: $p \notin \mathcal{R}_X$. $\square$

Theorem 6.3 (Necessity for compact sets). *Let $X \subseteq \mathbb{H}^n$ be compact and $p \in \mathcal{R}_X \setminus \widehat{X}$. Then the deepest free horoball $\mathcal{H}^*$ is horospherically defective and contains p . Necessity holds.*

Proof. By Lemma 6.1, $\pi \in \text{conv}(K^*)$. Were K^* convex, $\text{conv}(K^*) = K^*$, so $\pi \in K^*$, and Lemma 6.2 gives $p \notin \mathcal{R}_X$ — contrary to hypothesis. So K^* is non-convex: $\mathcal{H}^*$ is horospherically defective, and contains p as $\delta^* > 0$. $\square$

6.2 The non-compact case: the moving contact diagram

When X is non-compact the feet of a drift may escape to the ideal boundary, so the contact set K^* undercounts. The missing data is the perpendicular *offset* of each asymptotic ray (Example 12.15 in Part II shows the direction alone is insufficient), and the right object is a contact diagram whose asymptotic sites *move* with the height.

Definition 6.4 (Asymptotic rays and the offset set). *Fix a direction $\eta \in \partial_\infty S = S^{n-2}$ and write $\eta^\perp \cong \mathbb{E}^{n-2}$. The offset set of X in direction η is*

$$B_\eta := \left\{ b \in \eta^\perp : \exists x_k = (\beta_k, s_k) \in X, \langle \beta_k, \eta \rangle \rightarrow +\infty, \beta_k - \langle \beta_k, \eta \rangle \eta \rightarrow b, s_k \rightarrow 1 \right\},$$

the perpendicular positions of the lines along which X asymptotes to the horosphere in direction η . A maximal free horoball is ray-defective in direction η if B_η is non-convex in $\eta^\perp$. The full asymptotic-contact datum is K^ together with the family $(B_\eta)_\eta$ — points and oriented lines of S .*

We treat first the single direction with a common profile, where the offset reduction is exact. Suppose X approaches the horosphere only in direction $\eta = e_1$, along ray-graphs of a common profile,

$$\ell_b = \{(\chi \eta + b, 1 - \psi(\chi)) : \chi \geq \chi_0\} \subseteq X \quad (b \in B_\eta), \quad X \setminus \{z > 1 - \nu\} \subseteq \{z \leq 1 - \nu\},$$

with ψ smooth, positive, convex, decreasing to 0 (e.g. $\psi = \text{sech}$), and write a centre as $a = (a_1 \eta + a', t)$, $a' \in \eta^\perp$.

Lemma 6.5 (Feet see only the offset). *For bounded a_1, a' and $t \rightarrow \infty$,*

$$\cosh d(a, \ell_b) = \Phi(a_1, t) + \frac{|a' - b|^2}{2t} (1 + o(1)),$$

where $\Phi(a_1, t) = \min_\chi \frac{(a_1 - \chi)^2 + (t - 1 + \psi(\chi))^2}{2t(1 - \psi(\chi))} + 1$ is independent of the offset b and of the ray. Hence for two rays $d(a, \ell_{b_1}) \leq d(a, \ell_{b_2})$ according as $|a' - b_1| \leq |a' - b_2|$, and a is medial through them iff $|a' - b_1| = |a' - b_2|$.

Proof. The offset enters $\cosh d(a, \ell_b) = \min_\chi \frac{N(\chi) + |a' - b|^2}{2t(1 - \psi(\chi))} + 1$ only through the constant $K := |a' - b|^2$ added to $N(\chi) = (a_1 - \chi)^2 + (t - 1 + \psi)^2$. By the envelope theorem, $\frac{d}{dK} \min_\chi(\cdot) = \frac{1}{2t(1 - \psi(\chi^*(K)))}$. The minimiser χ^* lies at arclength $\sim \frac{2}{\lambda} \log t \rightarrow \infty$ (balancing $\frac{t\psi}{2}$ against $\frac{(\chi - a_1)^2}{2t}$), where $\psi(\chi^*) \rightarrow 0$, so $\frac{d}{dK} \min = \frac{1}{2t}(1 + o(1))$ uniformly for bounded K . Integrating from $K = 0$ gives the claim; the a_1 - and t -dependent (but offset-free) part is Φ , which cancels in any comparison of two rays. $\square$

Theorem 6.6 (Single-direction defect = Motzkin one dimension down). *Under the standing hypotheses, with the bulk uniformly below the horosphere, a high centre $a = (a_1 \eta + a', t)$ has its feet on the rays whose offsets are Euclidean-nearest to a' in $\eta^\perp$; therefore a is medial iff a' has ≥ 2 nearest points in B_η . Consequently reconstruction near the horoball occurs iff B_η is non-convex in $\eta^\perp \cong \mathbb{E}^{n-2}$ (Motzkin), the medial offsets being exactly its non-Chebyshev points.*

Proof. For t large the bulk ($z \leq 1 - \nu$) is farther from a than any ray (the leading term $\frac{t}{2(1-\psi)}$ is minimised on the rays, where $\psi \rightarrow 0$), so feet lie on rays; by Lemma 6.5 the nearest ray is the offset-nearest, and a is medial iff a' is non-Chebyshev for B_η . If B_η is non-convex, Motzkin (Theorem 4.1) supplies such an a' , and running a_1, t up the corresponding medial family reconstructs an open set (as in Proposition 12.16); if B_η is convex it is Chebyshev, every centre has a unique foot, and no medial point arises near the horoball. $\square$

Completing the square in Lemma 6.5 unifies all configurations — several directions, unequal profiles, mixed attained and asymptotic contacts — into one Euclidean Motzkin problem with *moving* contacts. With $\psi(\chi) \sim e^{-\lambda\chi}$ the minimising arclength is $\chi^* \sim \frac{2}{\lambda} \log t$, and

$$\cosh d(a, \ell_{\eta, \lambda, b}) = 1 + \frac{(t-1)^2}{2t} + \frac{|\alpha - P_{\eta, \lambda, b}(t)|^2}{2t} + o(1/t), \quad P_{\eta, \lambda, b}(t) := \frac{2}{\lambda}(\log t)\eta + b,$$

the same form as for an attained contact (with $P \equiv \kappa$). So at height t every contact presents as a point of $\mathbb{R}^{n-1}$ — attained ones fixed, asymptotic ones escaping along η — and a high centre $a = (\alpha, t)$ is medial iff α is non-Chebyshev for the moving set

$$K(t) := K^* \cup \{P_{\eta, \lambda, b}(t)\} \subseteq \mathbb{R}^{n-1}.$$

(This is the corrected form of the boundary-envelope idea discussed in §12: not a diagram on the conformal sphere, on which all asymptotic rays collapse to one pole, but an ordinary Voronoi/Motzkin diagram in $\mathbb{R}^{n-1}$ whose sites move.) Sufficiency is then immediate and gives the whole horoball, not a swept subregion.

Theorem 6.7 (Defect $\Rightarrow$ whole horoball). *Suppose the moving contact set $K(t)$ is non-convex in $\mathbb{R}^{n-1}$ for all large t (with the bulk uniformly below the horosphere). Then $\{z > 1\} \subseteq \mathcal{R}_X$.*

Proof. By Motzkin, $K(t)$ has a non-Chebyshev point $\alpha(t)$ — a medial centre $a(t) = (\alpha(t), t)$ with ≥ 2 equidistant nearest contacts, hence ≥ 2 feet, the ball $B(a(t), d(a(t)))$ free. For any fixed $p = (\pi, z_p)$ with $z_p > 1$, the centres have $|\alpha(t)| = O(\log t) = o(t)$, so

$$\cosh d(p, a(t)) = 1 + \frac{|\pi - \alpha(t)|^2 + (z_p - t)^2}{2z_p t} \sim \frac{t}{2z_p}, \quad \cosh d(a(t)) \sim \frac{t}{2},$$

and since $z_p > 1$ the former is the smaller for large t : $p \in B(a(t), d(a(t)))$. As $p \in \{z > 1\}$ was arbitrary, $\{z > 1\} \subseteq \mathcal{R}_X$. $\square$

For necessity we need the converse — that two feet witness non-convexity of $K(t)$ — and the only thing to rule out is the two feet lying on one ray. No-collapse, applied to a single ray, does exactly that.

Proposition 6.8 (No collapse on a single ray). *An asymptotic ray $\ell_{\eta, \lambda, b} = \{(\chi\eta + b, 1 - \psi(\chi))\}$ with convex tail profile ψ is a graph contact in the sense of Theorem 4.3 (height deficit $\phi = \psi$, convex on the tail). Hence a high centre has a unique foot on it: the minimiser of $F_t(\chi) = |\alpha - (\chi\eta + b)|^2 + t^2\psi(\chi)$ is unique, its second derivative $2 + t^2\psi'' > 0$. Consequently a centre with two feet has them on two distinct rays (or a ray and an attained contact), and these present as two distinct moving sites: $P_{\eta_1, \lambda_1, b_1}(t) \neq P_{\eta_2, \lambda_2, b_2}(t)$ for large t , whether the rays differ in direction, rate, or offset. So two feet force $K(t)$ non-convex.*

One subtlety makes the moving picture correct where a naive limit fails: the site a ray presents must be read at *finite* height. The site of a ray r at height t is the horizontal position of its foot-arclength,

$$P_r(t) = \chi^*(t)\eta_r + b_r(\chi^*(t)), \quad \chi^*(t) \sim \frac{2}{\lambda_r} \log t,$$

where $b_r(\chi)$ is the offset at arclength χ . For a straight ray this is the earlier $\frac{2}{\lambda}(\log t)\eta + b$; but the offset must be taken at $\chi^*(t)$, not in the limit $\chi \rightarrow \infty$. The distinction is decisive:

Example 6.9 (Converging ridges: the limit offset is the wrong invariant). In $\mathbb{H}^3$ take the two ridges $\{(\chi, \pm c/\chi, 1 - \psi(\chi))\}$, whose offsets $\pm c/\chi \rightarrow 0$. Both have the *same* limit offset 0, so the limit-offset set $B_\eta = \{0\}$ is convex and would predict no reconstruction. Yet the pair reconstructs the whole horoball (verified numerically: two genuine feet whose separation is stable under refinement, ≈ 0.206 at $t = 400$ across an eightfold mesh refinement, against a sheet's single foot which collapses to the mesh). The finite-height sites resolve it: at height t the ridges sit at $P_\pm(t) = \chi^*(t)\eta \pm c/\chi^*(t)$, *distinct* for every finite t (the foot-offset equals c/χ^* to four places: 0.166 at $t = 50$, 0.070 at $t = 5000$). So $K(t)$ is non-convex at every height even though its two sites coalesce as $t \rightarrow \infty$. Hence uniform separation of the limiting rays is *not* necessary for reconstruction: distinctness at each finite t is what matters, and that is exactly what Proposition 6.8 supplies.

With no-collapse on a ray and the finite-height site, necessity now closes: a reconstructing drift's two feet land on two distinct sites, so $K(t)$ is non-convex at every large height. No uniform separation of the limiting rays is needed — and, as Example 6.9 shows, none is available in general.

Theorem 6.10 (Necessity for graph-type sets). *Let $X \subseteq \mathbb{H}^n$ be graph-type near the horosphere (height deficit $\phi \geq 0$ locally semiconvex), and let $p \in \mathcal{R}_X \setminus \widehat{X}$ lie in the maximal free horoball $\{z > 1\}$ with $z_p > 1$. Then the finite-height site set $K(t) = K^* \cup \{P_r(t)\}_r$ is non-convex for all large t ; the horoball is defective.*

Proof. Since $p \notin \widehat{X}$, the trichotomy (Theorem 3.8) together with rigidity and screening (Proposition 3.3, Lemma 6.2: a unique-foot inflation escapes to the ideal boundary and cannot reconstruct) leaves only the drift branch — medial centres $a_k = (\alpha_k, t_k) \in M_X$, $t_k \rightarrow \infty$, each with ≥ 2 feet, the free balls converging to $\{z > 1\} \ni p$. By Lemma 4.2 every foot ascends, $s_k \rightarrow 1$. Each foot is therefore either bounded — an attained contact in K^* — or escapes along a ray (maximal convex-profile piece of X), where by Proposition 6.8 it is the *unique* foot on that piece. So the two feet of a_k sit on two *distinct* sites: two attained contacts, two ray-pieces, or one of each. Distinct sites are distinct points of $K(t)$ at the height in question — attained ones fixed, ray ones at $P_r(t) = \chi^*(t)\eta_r + b_r(\chi^*(t))$, separated because the pieces are disjoint escaping graphs with distinct foot-arclength positions. Hence $K(t)$ is non-convex. The contacts and pieces realising the two feet are t -independent features of X , so $K(t)$ is non-convex for all large t ; by Motzkin the horoball is defective, and by Theorem 6.7 it reconstructs its whole interior, consistently containing p . $\square$

7 The reconstruction theorem

Both inclusions are now in hand. We state the $\mathbb{H}^n$ characterisation in the moving-diagram form that covers compact and non-compact sets uniformly, then specialise to the compact case where the statement is cleanest.

Theorem 7.1 (Moving-contact defect; graph-type X). *For graph-type $X \subseteq \mathbb{H}^n$ a maximal free horoball $\mathcal{H}$ contributes to $\mathcal{R}_X$ if and only if the finite-height site set $K(t) = K^* \cup \{P_r(t)\}_r$ is non-convex in $\mathbb{R}^{n-1}$ for all large t ; the non-Chebyshev points of $K(t)$ are the medial offsets. Sufficiency is Theorem 6.7, necessity Theorem 6.10. This single Euclidean diagram subsumes K^* non-convex (compact case), B_η non-convex (Theorem 6.6), ≥ 2 escape directions, unequal-rate domination, and converging offsets (Example 6.9).*

Corollary 7.2 (The compact case, closed). *For compact graph-type $X \subseteq \mathbb{H}^n$,*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\mathcal{H} \text{ maximal free, horospherically defective}\}.$$

Conjecture 11.4 is a theorem in this class.

Proof. $\supseteq$: the first term is [2, Thm. 3.1], the second is Theorem 5.3. $\subseteq$: every $p \in \mathcal{R}_X \setminus \widehat{X}$ lies in its deepest horoball, horospherically defective, by Theorem 6.3 (proved in §6.1 via the deepest-shadow lemma — no drift needed). Since $K_\infty = \emptyset$ for compact X , extended defect coincides with horospherical. $\square$

This is Theorem 1.1. The reconstruction of $\mathbb{H}^n$ is thus complete for every graph-type set, in every dimension, compact or not: the hull term $\widehat{X} \setminus X$ is reconstructed by completeness alone [2, Thm. 3.1], and outside the hull a point is reconstructed iff it lies in a horospherically defective maximal free horoball, defect being measured by non-convexity of the (finite-height, moving) contact set. What lies strictly beyond the method is the non-graph regime — where the contacts do not decompose into convex-profile pieces — and the move off $\mathbb{H}^n$, where horospheres curve (§13).

Part II. Genesis, examples, and extensions

8 Bitangency, splitness, and the corrected notion

This part records how the statement of Theorem 1.1 was arrived at: the working notions that failed, the counterexample that forced the correction, the worked examples, and the routes to necessity that the deepest-shadow and moving-diagram arguments later subsumed. None of it is needed for Part I, but it explains every choice made there and isolates what remains open.

The starting point of [2] was the analogue of Białożyty's defectiveness, measured against the horoball hull. We first recall why it must be replaced.

Definition 8.1 (Bitangent; defective). *A maximal free horoball $\mathcal{H}_\gamma$ is bitangent if $S_\gamma \cap X$ contains at least two points, and defective (the notion of [2, Def. 2.5]) if $(S_\gamma \cap \widehat{X}) \setminus X \neq \emptyset$.*

Remark 8.2 (Why defectiveness must be replaced). For $X = \{x_1, x_2\} \subset \mathbb{H}^2$ one has $\widehat{X} = X$: in the upper half-plane model with $x_{1,2} = (\pm s, 1)$, every point $z = (u, v) \notin X$ with $v \leq 1$ lies in the free horoball tangent to the ideal boundary at $(u, 0)$ of Euclidean radius $\rho = \frac{1}{2}(\min_\pm(u \mp s)^2 + 1)$, and every point with $v > 1$ lies in the free horoball $\{y > 1\}$. (This corrects Remark 2.6 of [2]: the horoball hull does not even contain the geodesic segment $[x_1, x_2]$, whose interior dips into $\{y > 1\}$.) Consequently no maximal free horoball is defective, although $\mathcal{R}_X = \{y > 1\} \cup B_0 \neq \emptyset$, where B_0 is the horoball tangent at 0 through both points: Conjecture 5.1 of [2] fails as stated. Both horoballs constituting $\mathcal{R}_X$ are bitangent, which motivates Definition 8.1(a) and the revision below. We do not reproduce the full computation here, the original conjecture having been a working formulation.

Conjecture 8.3 (Revised necessity-and-sufficiency). *Let M be Hadamard and X closed. Then*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \left\{ \mathcal{H}_\gamma : \mathcal{H}_\gamma \text{ maximal free, split (Def. 9.2) or defective} \right\}.$$

These two notions, and the *split* refinement introduced to repair them, organise the rest of the genesis. We compare them, prove the split sufficiency theorem that the hull collapse forced, and then exhibit the dimension-three counterexample that retired splitness in favour of horospherical defect.

The two notions are incomparable in general; they agree for compact X in $\mathbb{R}^n$.

Proposition 8.4. (a) (**Bitangent $\not\Rightarrow$ defective, Hadamard**) *For X a two-point set in $\mathbb{H}^2$ (Remark 8.2), the two bitangent maximal free horoballs are not defective, since $\widehat{X} = X$.*

(b) (**Defective $\not\Rightarrow$ bitangent, already in $\mathbb{R}^n$**) *For $X = \text{graph}(e^x) \cup \{(5, 0)\} \subseteq \mathbb{R}^2$, the maximal free half-space $\{y < 0\}$ has the single attained contact $(5, 0)$, yet it is defective: points $(3 - \theta, 0)$, $\theta \geq 0$ small, lie in $\overline{\text{conv}} X \setminus X$ on the boundary line. Mass escaping to infinity in the hull can even produce defective half-spaces whose witness is unreachable by contact sequences: for $X = \{(5, 0)\} \cup \{(-k^2, -k) : k \in \mathbb{N}\}$, the half-space $\{y < -0\}$ ¹ is defective with witnesses $(5 - c, 0)$, $c \in (0, C]$, obtained from convex combinations with weights $\sim 1/k$ on $(-k^2, -k)$, while every sequence in X with $y \rightarrow 0$ is eventually constant equal to $(5, 0)$.*

(c) (**Characterisation for compact X in $\mathbb{R}^n$**) *If $X \subseteq \mathbb{R}^n$ is compact, a maximal free half-space $L^+ = \{\langle v, \cdot \rangle > c\}$, $c = \max_X \langle v, \cdot \rangle$, is defective if and only if the slice $\partial L^+ \cap X$ is not convex. In particular bitangency ($\#(\partial L^+ \cap X) \geq 2$) is necessary but not sufficient for defectiveness: a segment has a convex slice (cf. Example 9.1).*

Proof. (a) Remark 8.2. (b) The witness claims follow from computing limits of convex combinations as indicated; closedness of each X is immediate, and freeness and maximality of the half-spaces follow since $\inf$ is 0, attained at $(5, 0)$ in both cases. (c) ($\Leftarrow$) If the slice is not convex, pick $x_1, x_2 \in \partial L^+ \cap X$ and $z \in [x_1, x_2] \setminus X$; then $z \in \text{conv}(X \cap \partial L^+) \setminus X \subseteq (\overline{\text{conv}} X \cap \partial L^+) \setminus X$ is a witness. ($\Rightarrow$) For compact X , $\text{conv} X$ is compact and $\overline{\text{conv}} X \cap \partial L^+ = \text{conv}(X \cap \partial L^+)$: writing $w = \sum \lambda_i x_i$ with $x_i \in X$, $\langle v, w \rangle = c$ and $\langle v, x_i \rangle \leq c$ force $\langle v, x_i \rangle = c$ whenever $\lambda_i > 0$. A defective witness $w \notin X$ then lies in $\text{conv}(X \cap \partial L^+) \setminus (X \cap \partial L^+)$, so the slice differs from its convex hull: it is not convex. $\square$

Remark 8.5. Proposition 8.4(b) shows that Conjecture 8.3 genuinely needs the disjunction: defective horoballs cover reconstruction produced by asymptotic contacts (feet escaping to infinity, Remark 3.7), bitangent ones cover the hull-collapse regime where defectiveness is blind. Whether on a Hadamard manifold the drift branch with escaping feet always lands in a *defective* horoball is open; a robust common generalisation would count *contact ends* (limits of contact sequences in $M \cup \partial_\infty M$), but example (b) second part shows that even this undercounts the defective family in the presence of escaping deep mass. We leave the right asymptotic notion open (Question 14.2).

9 Sufficiency: split horoballs, and the failure of bare bitangency

Example 9.1 (Bitangency does not imply reconstruction). In the upper half-plane model of $\mathbb{H}^2$ let $X := \{(u, 1) : -1 \leq u \leq 1\}$, a compact horocyclic arc, and let γ be the upward vertical ray,

¹Precisely: the maximal free half-space in direction $-e_2$ through the contact $(5, 0)$.

so that $b_\gamma(x, y) = -\log y$ and $c_\gamma = 0$. The maximal free horoball $\mathcal{H}_\gamma = \{y > 1\}$ has contact set $S_\gamma \cap X = X$: it is bitangent with a continuum of contacts. Nevertheless $M_X = \emptyset$: for $(a, T) \notin X$,

$$\cosh d((a, T), (u, 1)) = 1 + \frac{(u - a)^2 + (T - 1)^2}{2T}$$

is strictly increasing in $(u - a)^2$, so the nearest point of the arc is the unique point with $u = \min(\max(a, -1), 1)$. Hence $\mathcal{R}_X = \emptyset \not\supseteq \mathcal{H}_\gamma$: *bitangent sufficiency is false*, and the second disjunct of Conjecture 8.3 must be the split family of Definition 9.2. The Euclidean shadow (X a segment in $\mathbb{R}^2$) is handled correctly by Theorem 6 of [1] because the boundary slice is convex and the half-space is not defective (Proposition 8.4(c)): what reconstruction detects is a *gap* in the contact set, not multiplicity.

Definition 9.2 (Split horoball). *A maximal free horoball $\mathcal{H}_\gamma$ is split at scale $\eta > 0$ if*

$$X \cap \{b_\gamma \leq c_\gamma + \eta\} = F_1 \cup F_2, \quad \text{dist}(F_1, F_2) > 0, \quad F_i \cap S_\gamma \neq \emptyset \quad (i = 1, 2),$$

with F_1, F_2 closed and nonempty.

Remark 9.3. A split horoball is bitangent. In $\mathbb{R}^n$ a split half-space is defective: choose contacts $x_i \in F_i \cap \partial L^+$ and bisect repeatedly, always replacing the endpoint lying in the same cluster as the midpoint; the pair distance halves, so after finitely many steps it drops below $2 \text{dist}(F_1, F_2)$, at which point the midpoint can lie in neither cluster, hence not in X — a witness. Thus splitness is invisible in the Euclidean statement; on a Hadamard manifold it is the part of defectiveness that survives the collapse of the hull (Remark 8.2).

Lemma 9.4 (Pure descent from a contact). *Let $\mathcal{H}_\gamma$ be maximal free, normalised $c_\gamma = 0$, let $x_0 \in S_\gamma \cap X$, and let σ_0 be the asymptotic ray from x_0 to $\xi = \gamma(\infty)$. Then for all $t > 0$: $b_\gamma(\sigma_0(t)) = -t$, $d(\sigma_0(t)) = t$, and $m(\sigma_0(t)) = \{x_0\}$.*

Proof. By (H3), $b_\gamma \circ \sigma_0$ has slope -1 and vanishes at $t = 0$. For $x \in X$ the 1-Lipschitz bound gives $d(x, \sigma_0(t)) \geq b_\gamma(x) - b_\gamma(\sigma_0(t)) \geq t$, while $d(x_0, \sigma_0(t)) = t$; hence $d(\sigma_0(t)) = t$ with x_0 a foot. If x is any foot, equality $d(x, \sigma_0(t)) = t$ forces $b_\gamma(x) = 0$ and unit-rate descent of b_γ along the segment from x to $\sigma_0(t)$; with $|\nabla b_\gamma| \equiv 1$ (H1), the unit-speed segment satisfies $\dot{c} = -\nabla b_\gamma(c)$, so it is an integral curve of the locally Lipschitz field $-\nabla b_\gamma$ through $\sigma_0(t)$, as is σ_0 ; uniqueness of integral curves gives $x = \sigma_0(0) = x_0$ (cf. Step 1 of [2, Thm. 4.1]). $\square$

Theorem 9.5 (Split sufficiency). *Let M be Hadamard, X closed, and $\mathcal{H}_\gamma$ split at scale η with clusters F_1, F_2 . Choose contacts $x_i \in F_i \cap S_\gamma$, let σ_i be the asymptotic rays from x_i to $\xi = \gamma(\infty)$, and set $L_\infty := \lim_{t \rightarrow \infty} d(\sigma_1(t), \sigma_2(t))$. If $\eta > L_\infty/2$, then $\mathcal{H}_\gamma \subseteq \mathcal{R}_X$.*

Proof. Normalise $c_\gamma = 0$ and write $b := b_\gamma$; by (H3), normalised at the contacts, $b_{\sigma_1} = b_{\sigma_2} = b$. The function $f(t) := d(\sigma_1(t), \sigma_2(t))$ is convex and bounded (the rays are asymptotic to the same ideal point), hence non-increasing, with $f(t) \downarrow L_\infty$. Fix t with $f(t) \leq 2\eta$ and let $J_t := [\sigma_1(t), \sigma_2(t)]$ be the geodesic segment.

Step 1: all feet over J_t lie in $F_1 \cup F_2$. Let $z \in J_t$ and $x \in m(z)$. Convexity of b along J_t gives $b(z) \leq \max_i b(\sigma_i(t)) = -t$. Since z lies on a segment of length $f(t)$, there is an i with $d(z, \sigma_i(t)) \leq f(t)/2$, whence $d(z) \leq d(z, x_i) \leq d(z, \sigma_i(t)) + d(\sigma_i(t), x_i) \leq t + f(t)/2$. The Lipschitz bound then yields $b(x) \leq b(z) + d(z, x) = b(z) + d(z) \leq f(t)/2 \leq \eta$, so $x \in X \cap \{b \leq \eta\} = F_1 \cup F_2$.

Step 2: a medial point $z_t \in J_t$ with $d(z_t) \geq t$. Set $A_i := \{z \in J_t : m(z) \subseteq F_i\}$. By Lemma 9.4, $\sigma_i(t) \in A_i$, so both sets are nonempty; they are disjoint because feet sets are nonempty and

$F_1 \cap F_2 = \emptyset$. Each A_i is relatively open: if $z_k \rightarrow z$ in J_t and each z_k has a foot $x_k \in m(z_k) \setminus F_1$, then $x_k \in F_2$ by Step 1; the x_k are bounded ($d(z_k) \leq t + f(t)/2$ with z_k in the compact J_t), so a subsequence converges to some x with $d(z, x) = \lim d(z_k, x_k) = \lim d(z_k) = d(z)$ and $x \in F_2 \cap X$ (both closed); thus $z \notin A_1$, and contrapositively A_1 is relatively open; symmetrically A_2 . Since J_t is connected, it is not the union of the two disjoint nonempty relatively open sets A_1, A_2 : there is $z_t \in J_t$ with feet in both clusters, hence $\#m(z_t) \geq 2$ and $z_t \in M_X$. Finally, for every $x \in X$, $d(x, z_t) \geq b(x) - b(z_t) \geq t$, so $d(z_t) \geq t$.

Step 3: the medial balls $B(z_t, d(z_t))$ exhaust $\mathcal{H}_\gamma$. Let $p \in \mathcal{H}_\gamma$ and $\beta := -b(p) > 0$. The quantity $d(p, \sigma_i(t)) - t$ is non-increasing in t with limit $b_{\sigma_i}(p) = -\beta$, so $d(p, \sigma_i(t)) \leq t - \beta + \varepsilon(t)$ with $\varepsilon(t) \downarrow 0$. Convexity of $d(p, \cdot)$ along J_t gives $d(p, z_t) \leq \max_i d(p, \sigma_i(t)) \leq t - \beta + \varepsilon(t)$. For t large enough that $\varepsilon(t) < \beta$ and $f(t) \leq 2\eta$,

$$d(p, z_t) < t \leq d(z_t),$$

so $p \in B(z_t, d(z_t)) \subseteq \mathcal{R}_X$. □

Corollary 9.6. *If the contact rays satisfy $L_\infty = 0$ — in particular whenever M has sectional curvature $\leq -\kappa^2 < 0$, since asymptotic geodesics then converge — every split horoball, at every scale $\eta > 0$, is contained in $\mathcal{R}_X$.*

Beyond split horoballs, the following triviality records the sufficiency of the limit objects produced by the necessity branches:

Proposition 9.7. *If a maximal free horoball $\mathcal{H}$ is exhausted by medial balls (for every compact $K \subseteq \mathcal{H}$ there is $a \in M_X$ with $K \subseteq B(a, d(a))$), then $\mathcal{H} \subseteq \mathcal{R}_X$. In particular the drift horoballs of Theorem 3.6 satisfy $\{b < s_\infty\} \subseteq \mathcal{R}_X$: their approximating centres are medial by hypothesis. For the single-foot escape horoballs of Theorem 3.4 no such conclusion is available — consistently with their single attained contact and with Example 9.1.*

Proof. Immediate: every point of $\mathcal{H}$ lies in some such compact K . □

Remark 9.8 (Where this leaves the conjecture). Sufficiency now reads: $\widehat{X} \setminus X$ always ([2, Thm. 3.1]); defective horoballs always ([2, Thm. 4.1]); split horoballs under $\eta > L_\infty/2$, hence unconditionally when $L_\infty = 0$ (Theorem 9.5, Corollary 9.6). Two gaps remain. On the sufficiency side: the regime $0 < 2\eta \leq L_\infty$, where the horoball is split only at scales below half the asymptotic separation of the contact rays (possible when those rays bound a flat half-strip); there neither Theorem 9.5 nor defectiveness applies. On the necessity side: the branches of §3.1 deliver maximal free horoballs with at least one attained contact (drift: subsequential limits of feet; ray escape: the inflation foot), and for reconstructible p these must be upgraded to *split-or-defective* (Question 14.1). By Proposition 3.3 the upgrade cannot come from a single inflation: the second contact cluster must be injected by equidistant motion of centres, for which the pincer of Theorem 9.5 is the prototype.

10 Progress on the split upgrade

Throughout this section $p \in X^c \setminus \widehat{X}$. Normalise all Busemann functions at a basepoint ($b_\xi(o) = 0$); then $(\xi, y) \mapsto b_\xi(y)$ is continuous on $\partial_\infty M \times M$ [4, II.8], so $c_\xi := \inf_X b_\xi \in [-\infty, \infty)$ is upper semicontinuous in ξ and the *depth function*

$$D(\xi) := c_\xi - b_\xi(p)$$

is upper semicontinuous on the compact $\partial_\infty M$; it is independent of the normalisation, positive somewhere precisely because $p \notin \widehat{X}$, and attains its maximum $\delta^* := \max_\xi D(\xi) > 0$ at some ξ^* .

Theorem 10.1 (Drift or asymptotic tangency). *Let $p \in X^c \setminus \widehat{X}$, let γ^* be a ray asymptotic to a depth-maximising ξ^* , and write $b^* := b_{\gamma^*}$, $c^* := c_{\gamma^*}$. Then either $T_{\delta^*/4}(p) = \infty$ (drift), or for all sufficiently large t the inflation from $\gamma^*(t)$ through any foot escapes, and in that case:*

- (a) $d(\gamma^*(t)) = t + c^* + o(1)$ (asymptotically exact free radii along the deepest axis);
- (b) every choice of feet $x_t \in m(\gamma^*(t))$ satisfies $b^*(x_t) \rightarrow c^*$ (asymptotic contacts of the deepest horoball);
- (c) every subsequential limit ξ' of the escape directions is itself depth-maximising: $D(\xi') = \delta^*$.

Proof. For $x \in X$ and every t , the non-increasing limit defining b^* gives $d(x, \gamma^*(t)) - t \geq b^*(x) \geq c^*$, so $d(\gamma^*(t)) \geq t + c^*$; set $\Delta_t := d(\gamma^*(t)) - t - c^* \geq 0$ and $\varepsilon(t) := d(p, \gamma^*(t)) - t - b^*(p) \downarrow 0$. The margin of p in the maximal free ball at $\gamma^*(t)$ is

$$\mu_t = d(\gamma^*(t)) - d(p, \gamma^*(t)) = \Delta_t + \delta^* - \varepsilon(t) > 0 \quad \text{for large } t.$$

If the inflations from $\gamma^*(t)$ are cut for an unbounded set of t , Lemma 3.2 (with $\varepsilon = \mu_t/4$) produces $a'_t \in M_X$ with $d(a'_t) \geq d(\gamma^*(t)) - \mu_t/4 \rightarrow \infty$ and $\mu(a'_t, p) \geq \mu_t/2 \geq (\delta^* - \varepsilon(t))/2 \geq \delta^*/4$ eventually: drift. Otherwise the inflation from $\gamma^*(t)$ escapes for all large t ; Theorem 3.4 places p at depth $\geq \mu_t$ in the maximal free horoball of the escape direction ξ_t , whence

$$\Delta_t + \delta^* - \varepsilon(t) = \mu_t \leq D(\xi_t) \leq \delta^*,$$

so $\Delta_t \leq \varepsilon(t) \rightarrow 0$, proving (a). For (b), $c^* \leq b^*(x_t) \leq d(x_t, \gamma^*(t)) - t = c^* + \Delta_t$. For (c), $D(\xi_t) \geq \mu_t \rightarrow \delta^*$ and upper semicontinuity give $D(\xi') \geq \delta^*$ for any subsequential limit, while $\leq \delta^*$ by maximality. $\square$

Remark 10.2. In the non-drift case the deepest horoball is asymptotically tangent along its own axis: the obstruction to drift is *exactly* an asymptotic-contact regime. The next example shows that this regime is, at least in the model case, the territory of defectiveness — the hull does *not* collapse against asymptotic strands.

Example 10.3 (Asymptotic strand: defective but not split). In the upper half-plane model of $\mathbb{H}^2$ let

$$X := \{x_1\} \cup \{(u, 1 - e^u) : u \leq -1\}, \quad x_1 := (0, 1),$$

a contact point together with a strand asymptotic to the horosphere $\{y = 1\}$ from below. For the upward ray γ ($b_\gamma = -\log y$, $c_\gamma = 0$), the maximal free horoball $\mathcal{H}_\gamma = \{y > 1\}$ has the single attained contact x_1 , so it is not split (a second cluster meeting S_γ does not exist). It is defective: $w := (-10, 1)$ is a trapped witness. Indeed, a free horoball containing w centred at ∞ is some $\{y > c\}$ with $c < 1$, which meets the strand (heights $\rightarrow 1$). A free horoball centred at $\xi \in \mathbb{R}$ is a Euclidean disk tangent at $(\xi, 0)$ of radius ρ ; containing w forces $(\xi + 10)^2 + 1 < 2\rho$, freeness against x_1 forces $2\rho \leq \xi^2 + 1$ (so $\xi < -5$), and then freeness against the strand point $(\xi, 1 - e^\xi)$ forces $2\rho \leq 1 - e^\xi < 1$ — contradicting $2\rho > (\xi + 10)^2 + 1 \geq 1$. Hence w lies in no free horoball: $w \in (S_\gamma \cap \widehat{X}) \setminus X$, and $\mathcal{H}_\gamma \subseteq \mathcal{R}_X$ by [2, Thm. 4.1]. Defectiveness thus covers the asymptotic-contact regime in which splitness is unavailable, complementing Example 9.1.

Proposition 10.4 (Split upgrade on $\mathbb{H}^2$: non-degenerate drift). *Let $M = \mathbb{H}^2$ and let p be in the drift case, with medial balls $B_k := B(a_k, r_k) \ni p$ (margins $\geq \delta$, $r_k \rightarrow \infty$) as in Theorem 3.6, whose feet pairs $q_k \neq q'_k$ subconverge to distinct contacts $x^* \neq x^{*'}$ of the limit horoball $\mathcal{H}^*$. Then $\mathcal{H}^*$ is split; consequently (Corollary 9.6) $\mathcal{H}^* \subseteq \mathcal{R}_X$, and p lies in a split maximal free horoball at depth $\geq \delta$.*

Proof. Work in the upper half-plane model, normalised so that the limit direction is $\xi = \infty$, $b(x, y) = -\log y$, and the limit level s_∞ is the horosphere $\{y = Y\}$; then $X \subseteq \{y \leq Y\}$ (Theorem 3.6(a)), the contacts are $x^* = (u_1, Y)$, $x^{*'} = (u_2, Y)$ with $G := u_2 - u_1 > 0$, and $b(x^*) = s_\infty = c_\xi$, attained. Hyperbolic balls are Euclidean disks, so each B_k is a Euclidean disk; by the exhaustion property of Theorem 3.6(b) the B_k eventually contain Euclidean disks of any prescribed radius inside $\{y > Y\}$, hence their Euclidean radii satisfy $R_k \rightarrow \infty$.

Clearance. The boundary circle of B_k passes through $q_k \rightarrow (u_1, Y)$ and $q'_k \rightarrow (u_2, Y)$, whose heights are $\leq Y$ (they lie in X). Let ℓ_k be the chord through them; $\ell_k \leq Y$ on the segment, and the lower boundary arc lies below ℓ_k , with sagitta at horizontal distance $\geq G/4$ from both feet at least $2\beta_k$, where $\beta_k := G^2/(128R_k) > 0$ (parabolic approximation of a circle of radius $R_k \gg G$; the feet positions converge to u_1, u_2 , so for large k the middle half $I := [u_1 + G/4, u_2 - G/4]$ of the positions lies strictly between them). Consequently, for large k , every point with position in I and height in $[Y - \beta_k, Y]$ lies strictly between the lower arc and the upper arc, i.e. in the open disk B_k ; freeness forbids X there.

Partition. In b -levels, heights $[Y - \beta_k, Y]$ are exactly $\{s_\infty \leq b \leq s_\infty + \eta_k\}$ with $\eta_k := -\log(1 - \beta_k/Y) > 0$, and points with $b \leq s_\infty + \eta_k$ have height $\geq Y - \beta_k$. Hence $X \cap \{b \leq s_\infty + \eta_k\}$ contains no point with position in I , and splits as $F_1 \cup F_2$ by position ($\leq u_1 + G/4$ versus $\geq u_2 - G/4$). Both are closed, contain the contacts x^* , $x^{*'}$ respectively, and have positive hyperbolic separation: any path joining them either stays below height H at horizontal cost $\geq (G/2)/H$, or climbs to height H at cost $\geq 2\log(H/Y)$, so $\text{dist}(F_1, F_2) \geq \min(G/(2Y), 2\log \frac{G}{4Y} + 2) > 0$. Thus $\mathcal{H}^*$ is split at scale η_k ; since $L_\infty = 0$ on $\mathbb{H}^2$, Theorem 9.5 applies at every scale. $\square$

Remark 10.5 (The remaining core). After Theorem 10.1 and Proposition 10.4, what remains of Question 14.1 is exactly:

- (i) *degenerate drift*: feet pairs merging ($d(q_k, q'_k) \rightarrow 0$) or escaping to infinity — the clearance scale β_k and the separation both degenerate with G ;
- (ii) *the non-drift case*: upgrading the asymptotic tangency of Theorem 10.1 to defectiveness, for which Example 10.3 is the model;
- (iii) *dimension ≥ 3* : the cleared region is a disk-shaped shadow in the horosphere $\cong \mathbb{R}^{n-1}$, whose complement is connected — contact chains may detour around it. This is not a mere proof-technique obstruction: §11 shows that for $n \geq 3$ the split-or-defective conjecture is *false*, and that splitness is a two-dimensional artefact. The remaining cases (i), (ii) are therefore moot for Conjecture 8.3 as stated; they survive only relative to the corrected notion of §11.

11 The conjecture fails in dimension three

The split refinement was forced by the collapse of the horoball hull (Remark 8.2); the two-point set in $\mathbb{H}^2$ recovered reconstruction through bitangency, two *separated* contacts. We now show that in $\mathbb{H}^3$ a single connected contact set already defeats Conjecture 8.3: splitness is a coincidence of dimension two.

Theorem 11.1 (A counterexample in $\mathbb{H}^3$). *Work in the upper half-space model $\mathbb{H}^3 = \{(x, y, z) : z > 0\}$, $\cosh d(p, q) = 1 + |p - q|^2 / (2z_p z_q)$, and let*

$$X = \{(\cos \theta, \sin \theta, 1) : \theta \in [0, 2\pi)\},$$

the unit circle on the horosphere $\{z = 1\}$. Write $R = \sqrt{x^2 + y^2}$ and $\mathcal{H}_0 = \{R^2 + (z - 1)^2 < 1\}$ (the open Euclidean ball of centre $(0, 0, 1)$ and radius 1, a free horoball centred at the ideal point 0). Then

- (a) $\widehat{X} = X$: *the horoball hull collapses to the circle;*
- (b) $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0$, *a non-empty open set; it is exactly the union of the two maximal free horoballs whose horospheres contain all of X ;*
- (c) *every maximal free horoball meets X in a connected set (either a single point or the whole circle); hence none is split, and by (a) none is defective;*
- (d) *consequently the right-hand side of Conjecture 8.3 is empty while $\mathcal{R}_X \neq \emptyset$. The conjecture is false for $n = 3$.*

Proof. The medial axis is the z -axis: for $a = (0, 0, h)$, $h > 0$, $\cosh d(a, (\cos \theta, \sin \theta, 1)) = 1 + \frac{1+(h-1)^2}{2h} = \frac{h^2+2}{2h}$ independently of θ , so all of X is equidistant and a is medial; any point off the axis has, by strict monotonicity of $\cosh d$ in $-2R \cos \theta$, a unique nearest point, so is not medial.

(b) A point $p = (x, y, z)$ lies in the medial ball $B(a, d(a))$ iff $\cosh d(p, a) < \cosh d(a)$, i.e. $1 + \frac{R^2+(z-h)^2}{2zh} < \frac{h^2+2}{2h}$. Clearing denominators ($2zh > 0$) and cancelling $-2zh$,

$$R^2 + z^2 - 2z + (1 - z)h^2 < 0. \quad (*)$$

If $z > 1$ the coefficient $(1 - z)$ is negative, so $(*)$ holds for h large: $\{z > 1\} \subseteq \mathcal{R}_X$. If $z = 1$, $(*)$ reads $R^2 < 1$. If $z < 1$, the left side is increasing in h^2 , with infimum $R^2 + z^2 - 2z$ as $h \rightarrow 0^+$, so $(*)$ is solvable iff $R^2 + (z - 1)^2 < 1$. Since off-axis points are not medial, these axis balls give all of $\mathcal{R}_X$; their union is $\{z > 1\} \cup \mathcal{H}_0$. The horospheres $\{z = 1\}$ (ideal point ∞) and $\partial\mathcal{H}_0$ (ideal point 0) both contain the entire circle.

(a) Each $q = (x_0, y_0, z_0) \notin X$ lies in a free open horoball: if $z_0 > 1$, in $\{z > 1\}$; if $R_0^2 + (z_0 - 1)^2 < 1$, in $\mathcal{H}_0$; otherwise $z_0 \leq 1$, $R_0 > 0$, and the horoball centred at the ideal point $(x_0, y_0, 0)$ — the Euclidean ball of centre (x_0, y_0, ρ) and radius ρ — is free iff $\rho \leq \frac{1}{2}((R_0 - 1)^2 + 1)$ and contains q iff $\rho > z_0/2$; these are compatible because $z_0 - 1 < (R_0 - 1)^2$ (strict, as $q \notin X$). Hence $M \setminus X$ is covered by free horoballs and $\widehat{X} = X$.

(c) A maximal free horoball is centred at an ideal point ξ . For $\xi = \infty$ it is $\{z > 1\}$, contact set the whole circle. For finite ξ of Euclidean norm σ it is the Euclidean ball of centre (ξ, ρ^*) , radius $\rho^* = \frac{1}{2}((\sigma - 1)^2 + 1)$, and the squared Euclidean distance from its centre to $(\cos \theta, \sin \theta, 1)$ equals $(\rho^* - 1)^2 + \sigma^2 - 2\sigma \cos \theta' + 1$ (θ' measured from ξ), minimised at the unique $\theta' = 0$ when $\sigma > 0$ — one contact — and constant when $\sigma = 0$ — the whole circle ($\mathcal{H}_0$). In every case the contact set is connected, so cannot be partitioned into two non-empty closed pieces at positive distance: no maximal free horoball is split. By (a), $\widehat{X} \setminus X = \emptyset$, so none is defective either.

(d) The right-hand side of Conjecture 8.3 is $(\widehat{X} \setminus X) \cup \bigcup \{\text{split or defective}\} = \emptyset$, whereas $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0 \neq \emptyset$. $\square$

Remark 11.2 (Splitness is a two-dimensional artefact). A horosphere S_γ of a Hadamard manifold is intrinsically flat, $S_\gamma \cong \mathbb{E}^{n-1}$. The reconstruction in Theorem 11.1 is driven by the contact set $K = S_\gamma \cap X$ failing to be *intrinsically convex* in S_γ : the circle’s flat convex hull is the disk, and the reconstructed points sit over the disk. In dimension $n = 2$ the horosphere is a line, $S_\gamma \cong \mathbb{E}^1$, where “convex hull strictly larger than K ” is the same as “ K has a gap”, i.e. K disconnects — which is precisely splitness. For $n - 1 \geq 2$ a connected set (the circle) can have strictly larger convex hull, so splitness (a connectivity property) is strictly weaker than the operative condition and misses the circle entirely. This is the exact mechanism by which Conjecture 8.3 fails.

Proposition 11.3 (The corrected notion passes the tests).

- (a) (Circle.) *In Theorem 11.1 the horospherically defective maximal free horoballs are exactly $\{z > 1\}$ and $\mathcal{H}_0$ (contact set the whole circle, flat hull the disk); their union is $\mathcal{R}_X$. The single-contact horoballs are not horospherically defective.*
- (b) (Euclidean reduction.) *For $M = \mathbb{R}^n$ and compact X , $\mathcal{H} = L^+$ is horospherically defective iff the slice $L \cap X$ is non-convex iff L is defective in the sense of [1] (Proposition 8.4(c)). Thus Definition 2.3 reduces to the hypothesis of Theorem 6.*
- (c) (Planar reduction.) *For $M = \mathbb{H}^2$ a horoball whose contact set is closed equals/refines splitness: a contact set $K \subset S \cong \mathbb{E}^1$ has $\text{hconv}(K) \neq K$ iff K has a gap, the on-the-horosphere shadow of the split condition (Definition 9.2).*

Proof. (a) Computed in the proof of Theorem 11.1(c): only $\xi \in \{0, \infty\}$ give multi-point (whole-circle) contact, with flat hull the disk $\neq$ circle; finite $\xi \neq 0$ give a single contact, flat hull a point. The union of the two is $\{z > 1\} \cup \mathcal{H}_0 = \mathcal{R}_X$ by (b) of that theorem. (b) For compact X the face $\overline{\text{conv}} X \cap L = \text{conv}(L \cap X)$, so the ambient defect witness $w \in (\overline{\text{conv}} X \cap L) \setminus X$ exists iff $\text{conv}(L \cap X) \neq L \cap X$; and on the hyperplane L the intrinsic and ambient convex hulls coincide. (c) On a line the closed convex hull of K is the smallest interval containing it, which differs from K exactly when K omits an interior point, i.e. has a gap. $\square$

Conjecture 11.4 (Corrected necessity-and-sufficiency). *Let M be Hadamard and X closed. Then*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \left\{ \mathcal{H}_\gamma : \mathcal{H}_\gamma \text{ maximal free, horospherically defective} \right\}.$$

Three caveats keep this honest. First, the strand (Example 10.3) shows that contacts can be *asymptotic* — X approaching S_γ without meeting it — so that K_γ undercounts; there $\mathcal{H}_\gamma$ is defective in the ambient sense but not horospherically defective, so the corrected union must be read as horospherically defective *or* defective, and the genuinely correct object is presumably the flat convex hull of the *asymptotic* contact set $\overline{\{x \in X : b_\gamma(x) \rightarrow c_\gamma\}}$ in a suitable horospherical compactification (Question 14.2). Second, for a horospherically defective $\mathcal{H}_\gamma$ it is the points lying *over* $\text{hconv}_\gamma(K_\gamma) \setminus X$ that should be reconstructed; whether this is the whole horoball (as for the circle and in $\mathbb{R}^n$) or a proper subregion in general needs the inflation analysis of §3 re-run with horospherical feet. Third, sufficiency ($\supseteq$) is now itself open in dimension ≥ 3 : the split sufficiency Theorem 9.5 must be upgraded to a horospherical-hull statement, the natural vehicle being a *wall* of medial centres swept over a triangulation of $\text{hconv}_\gamma(K_\gamma)$ rather than the single segment J_t of the planar pincer.

Corollary 11.5 (The circle, completed). *For the unit circle of Theorem 11.1, both maximal free horoballs with whole-circle contact — $\{z > 1\}$ and $\mathcal{H}_0$ — satisfy $\mathcal{H} \subseteq \mathcal{R}_X$, because the circle*

is non-convex on each of their (flat) horospheres. Hence $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0$ equals the union of the horospherically defective horoballs, and Conjecture 11.4 holds, as an equality, for this set. The example that refuted Conjecture 8.3 thus confirms its repair.

12 Further results and development notes

We collect here the results that are correct and informative but were superseded, as proof routes, by the deepest-shadow lemma and the moving diagram of Part I: the earlier non-merging route to necessity, the drift-or-hull reduction and its extended hull, the $\mathbb{H}^2$ non-compact analysis, and the offset/boundary-envelope development that led to the moving contact diagram. The rigidity remark that redirected the search opens the account.

Remark 12.1. Proposition 3.3 redirects the search for a necessity mechanism. Escaping inflations cannot start at medial centres, so a single inflation can never tag two feet on a limit horosphere; bitangency must instead be produced by moving centres *along equidistant loci* of two contact clusters — which is exactly what the pincer in the proof of Theorem 9.5 does, and what the medial families of the worked examples (e.g. the bisector family of a two-point set in $\mathbb{H}^2$) actually look like. An earlier draft of this note stated a bitangency theorem for escaping inflations from medial centres; the rigidity above shows its hypothesis is unsatisfiable.

12.1 The earlier necessity route: the non-merging estimate

Before the deepest-shadow lemma, necessity was approached through the feet of a drift directly: they ascend to the horosphere and, unless they merge, land on two distinct contacts, forcing non-convexity. The estimate below shows merging can only happen by collapse onto a single contact, which a graph contact never permits. For compact X this gives necessity for bounded drifts; the deepest-shadow lemma of Part I later removed the drift hypothesis entirely.

Proposition 12.2 (Contact feet force a defect). *If some medial centre a reconstructing p has two feet $q \neq q'$ both lying in K , then K is non-convex and $\mathcal{H}$ is horospherically defective. No further hypotheses.*

Proof. q, q' are nearest points of X to $a = (\alpha, t)$, both in $K \subseteq \{z = 1\}$. By monotonicity at height 1, their positions κ, κ' are Euclidean-nearest to α in K , at a common distance; since $\kappa \neq \kappa'$, the point α has two nearest points in K . So K is not Chebyshev, hence by Theorem 4.1 not convex. $\square$

Lemma 12.3 (Ascent). *Let $a_k = (\alpha_k, t_k)$ be medial with $t_k \rightarrow \infty$ and $\alpha_k \rightarrow \alpha_\infty$, and let $q_k = (\beta_k, s_k) \in m(a_k)$ have bounded positions β_k . Then $1 - s_k = O(t_k^{-2})$; in particular every subsequential limit of q_k lies in K , and $h_k := (1 - s_k)t_k^2$ is bounded.*

Proof. Fix a contact $k_0 = (\kappa_0, 1) \in K$. As q_k is a nearest point, $\cosh d(a_k, q_k) \leq \cosh d(a_k, k_0)$, i.e.

$$\frac{|\alpha_k - \beta_k|^2 + (t_k - s_k)^2}{s_k} \leq |\alpha_k - \kappa_0|^2 + (t_k - 1)^2.$$

Dropping the non-negative position term and using $\frac{(t_k - s_k)^2}{s_k} - (t_k - 1)^2 = \frac{(1 - s_k)(t_k^2 - s_k)}{s_k}$ gives $\frac{(1 - s_k)(t_k^2 - s_k)}{s_k} \leq |\alpha_k - \kappa_0|^2 \leq C$, whence $(1 - s_k)t_k^2 \leq C'$ for large k . Closedness of X then places limits of q_k in $X \cap \{z = 1\} = K$. $\square$

Theorem 12.4 (Non-merging estimate). *Let $a_k = (\alpha_k, t_k)$ be medial centres of a drift reconstructing $p \in \mathcal{H}$ ($t_k \rightarrow \infty$, $\alpha_k \rightarrow \alpha_\infty$, margins bounded below), with distinct feet $q_k \neq q'_k$ of bounded position. Passing to a subsequence, let $q_k \rightarrow q_* = (\kappa_*, 1)$ and $q'_k \rightarrow q'_* = (\kappa'_*, 1)$ in K (Lemma 12.3). Then:*

- (a) *the heights ascend sharply, $h_* = \lim(1 - s_k)t_k^2 = 0$ and likewise for q'_k ; both κ_*, κ'_* are Euclidean-nearest to α_∞ in K ;*
- (b) *(estimate) either $\kappa_* = \kappa'_*$ (collapse), or $\kappa_* \neq \kappa'_*$ and then*

$$d(q_k, q'_k) \rightarrow \operatorname{arccosh}\left(1 + \frac{1}{2}|\kappa_* - \kappa'_*|^2\right) > 0;$$

there is no intermediate decay rate;

- (c) *in the non-collapse case K has two distinct nearest points to α_∞ , so by Motzkin it is non-convex: $\mathcal{H}$ is horospherically defective and contains p . Necessity holds.*

Proof. Write $h_k = (1 - s_k)t_k^2$, $h'_k = (1 - s'_k)t_k^2$, bounded by Lemma 12.3; pass to limits $h_*, h'_* \geq 0$. For any contact $(\kappa, 1)$ the nearest-point inequality $\cosh d(a_k, q_k) \leq \cosh d(a_k, (\kappa, 1))$, multiplied by $2t_k$ and rearranged with $\frac{(t_k - s_k)^2}{s_k} - (t_k - 1)^2 = \frac{(1 - s_k)(t_k^2 - s_k)}{s_k} \rightarrow h_*$, passes to the limit as

$$|\alpha_\infty - \kappa_*|^2 + h_* \leq |\alpha_\infty - \kappa|^2 \quad (\forall \kappa \in K), \quad (\dagger)$$

and symmetrically $|\alpha_\infty - \kappa'_*|^2 + h'_* \leq |\alpha_\infty - \kappa|^2$. Putting $\kappa = \kappa'_*$ in the first and $\kappa = \kappa_*$ in the second and adding yields $h_* + h'_* \leq 0$, so $h_* = h'_* = 0$: this proves (a), and $(\dagger)$ becomes $|\alpha_\infty - \kappa_*| = |\alpha_\infty - \kappa'_*| = \min_{\kappa \in K} |\alpha_\infty - \kappa|$.

For (b), $d(q_k, q'_k) \rightarrow d(q_*, q'_*)$, and for two points at height 1, $\cosh d(q_*, q'_*) = 1 + \frac{1}{2}|\kappa_* - \kappa'_*|^2$. If $\kappa_* \neq \kappa'_*$ this is > 1 , the stated bound; the absence of intermediate rates is the dichotomy “equal or not”. For (c), distinct equidistant nearest points show K non-Chebyshev; Motzkin gives non-convexity, and p lies in $\mathcal{H} = \{z > 1\}$ by construction. $\square$

Remark 12.5 (The two ways out). The estimate leaves two ways for necessity to be threatened: *collapse* ($q_k, q'_k \rightarrow$ a single contact) and *escape* (q_k unbounded). The next section shows collapse never occurs for graph-type X (Theorem 4.3), so for bounded drifts the feet always converge to two distinct contacts and necessity holds outright; the sole genuine residue is escape, treated in §12.2 via the extended hull.

Remark 12.6 (Collapse does not occur). This is confirmed numerically: for $\phi = \chi^4$ (convex, degenerate) F_t has a single critical point for every off-axis α and all t — the root of $\chi + 2t^2\chi^3 = \alpha$, unique since the left side is strictly increasing — while for a double well $\phi = (\chi^2 - \varepsilon^2)^2$ the two feet lock onto $\pm\varepsilon$ (two separated contacts), never merging.

Corollary 12.7 (Necessity for bounded drifts; compact sets). *For graph-type $X \subseteq \mathbb{H}^n$, every $p \in \mathcal{R}_X \setminus \widehat{X}$ reconstructed by a drift with bounded feet lies in a horospherically defective horoball. In particular this holds for every compact X , whose feet are automatically bounded: necessity holds with no residue.*

Proof. Bounded feet ascend to contacts (Lemma 12.3); by Theorem 4.3 they cannot collapse, so by Theorem 12.4 they converge to two distinct contacts. Then K is non-Chebyshev, hence non-convex (Motzkin), and the horoball is horospherically defective and contains p . For compact X all feet lie in the bounded set X , so no drift can escape. $\square$

Remark 12.8 (What is proved, and what remains). Theorems 5.1–5.2 prove sufficiency for every horospherically defective horoball in $\mathbb{H}^n$ whose contacts are attained and isolated from the bulk; in particular the corrected conjecture’s $\supseteq$ holds, and Corollary 11.5 closes the circle counterexample. Three frontiers remain. (i) *Tangential bulk*. If X approaches the horosphere tangentially at a contact (mass $(\chi_j, 1 - \eta_j)$ with $\eta_j = o(|\chi_j - \kappa|)$), a fixed medial direction can be shadowed; one expects reconstruction to survive through a *moving* medial direction, but the clean Motzkin argument no longer applies verbatim. (ii) *Curved horospheres*. Outside $\mathbb{H}^n$ the horosphere need not be flat, and even “intrinsically convex” must be reread (the induced metric is a Hadamard metric only under curvature pinching); Motzkin’s theorem must be replaced by its Chebyshev analogue in the relevant Alexandrov class. (iii) *Necessity*. The reverse inclusion is the horospherical form of the upgrade (Question 14.1): a reconstructible $p \notin \hat{X}$ sits in a medial ball whose two feet, followed along the structure trichotomy of §3.1, must be shown to limit onto a *non-convex* contact set. By Motzkin this is equivalent to the limit contacts failing to be Chebyshev — exactly the multiplicity that the two feet $q \neq q'$ encode, provided they neither merge nor are filled in by X in the limit. This reduces necessity to a quantitative non-merging estimate, the higher-dimensional descendant of the planar Proposition 10.4.

12.2 Drift-or-hull and the extended horospherical hull

The reduction of $\subseteq$ to drift, and the extended hull that governs the escaping (non-compact) tail, were the framework before the moving diagram. The extended hull adjoins to the attained contacts the *ideal* contacts — directions in which X asymptotes to the horosphere — and recovers both the circle (a horospherical defect) and the strand (an ambient, asymptotic defect). It captures the right shape of the asymptotic contact set but, as Example 12.15 shows, not its location; that defect is what the finite-height moving site of Part I repairs.

Proposition 12.9 (Drift-or-hull, and its kernel). *Let $p \in \mathcal{R}_X \setminus \hat{X}$. Running the structure theorem (Theorem 3.8) along a free-horoball ray through p : if the inflations are cut, p drifts and Corollary 12.7 applies. The only alternative is that every free-horoball direction through p escapes; we claim this cannot persist for a reconstructible p , but the claim is not elementary. Two mechanisms force a defective horoball:*

- (a) (unbounded bisector) *the witness feet $q_0 \neq q'_0$ span a complete totally geodesic bisector, carrying medial centres of radius $\rightarrow \infty$; when one such subfamily inherits a uniform margin from the depth of p , p drifts. The three-point set in $\mathbb{H}^2$ realises this — its circumcentre is a bounded local maximum of d , yet p near it drifts along the bisector rays, so “bounded medial $\Rightarrow$ hull” is false; what is true is the weaker drift-or-hull.*
- (b) (gap horoball) *for an arc with an opening (a “C”), the centre is reconstructed by a bounded medial ball yet lies in the maximal free horoball entering through the gap, whose horosphere meets X at the two gap edges — a 2-contact, horospherically defective horoball. Here necessity holds with no drift at all.*

The geodesic sphere is the boundary case: feet form a continuum that blocks every horoball, so $o \in \hat{X}$ (verified in the half-space model) and there is nothing to prove. Transporting the two witness feet onto a common free horosphere, against the rigidity of Proposition 3.3, looked like the residual content of $\subseteq$. It is not needed: §6.1 bypasses the flow altogether for compact X — the deepest free horoball through p is already defective (Theorem 6.3) — so the kernel is dissolved, not merely reduced.

Definition 12.10 (Extended horospherical hull). *Compactify the flat horosphere $S \cong \mathbb{R}^{n-1}$ to a closed disc $\bar{S} = \mathbb{R}^{n-1} \cup \partial_\infty S$. To the attained contacts $K \subseteq \mathbb{R}^{n-1}$ adjoin the ideal contacts*

$$K_\infty := \left\{ \eta \in \partial_\infty S : \exists x_j = (\chi_j, s_j) \in X, \frac{\chi_j}{|\chi_j|} \rightarrow \eta, |\chi_j| \rightarrow \infty, s_j \rightarrow 1 \right\},$$

the directions in which X asymptotes to the horosphere. Call $\mathcal{H}$ extended-defective if $\overline{\text{conv}}(K \cup K_\infty) \cap \mathbb{R}^{n-1} \neq K$ — the closed convex hull (in $\bar{S}$) of attained and ideal contacts strictly exceeds K .

Conjecture 12.11 (Unified necessity-and-sufficiency, $\mathbb{H}^n$). $\mathcal{R}_X = (\hat{X} \setminus X) \cup \bigcup \{ \mathcal{H} \text{ maximal free, extended-defective} \}$

Extended defect unifies the two mechanisms. *Circle* (Theorem 11.1): K the circle, $K_\infty = \emptyset$, $\text{conv} = \text{disc} \neq \text{circle}$ — extended-defective, as a horospherical defect. *Strand* (Example 10.3): $K = \{(0, 1)\}$, and the strand $\chi \rightarrow -\infty$ at height $\rightarrow 1$ gives $K_\infty = \{-\infty\}$, so $\text{conv}(\{0\} \cup \{-\infty\})$ is the ray $(-\infty, 0]$, whose finite part $\neq \{0\}$ — extended-defective, as an asymptotic (ambient) defect, with witness any $(-c, 1)$, $c > 0$ (matching $w = (-10, 1)$). Definition 2.3 is the case $K_\infty = \emptyset$; the ambient defect of [2, Thm. 4.1] is recovered when an interior witness is shielded, which $K_\infty \neq \emptyset$ encodes.

Proposition 12.12 (Compactified non-merging: one escaping foot). *In $\mathbb{H}^n$ let $\mathcal{H} = \{z > 1\}$ be maximal free and let p be reconstructed by a drift $a_k = (\alpha_k, t_k)$, $t_k \rightarrow \infty$, with feet $q_k \rightarrow q_* = (\kappa_*, 1) \in K$ (bounded) and $q'_k = (\beta'_k, s'_k)$ with $|\beta'_k| \rightarrow \infty$ (escaping); the latter ascends, $s'_k \rightarrow 1$, by Lemma 4.2. Then $\eta := \lim \beta'_k / |\beta'_k| \in K_\infty$, the points $\kappa_* \in \mathbb{R}^{n-1}$ and $\eta \in \partial_\infty S$ are distinct in $\bar{S}$, and the closed ray $\kappa_* + \mathbb{R}_{\geq 0} \eta \subseteq \overline{\text{conv}}(K \cup K_\infty)$ has interior points outside X . Hence $\mathcal{H}$ is extended-defective (Definition 12.10) and contains p : necessity holds.*

Proof. The escaping foot $q'_k \in X$ has position $\rightarrow \infty$ along η and height $\rightarrow 1$, so $\eta \in K_\infty$ by definition. In the disc $\bar{S}$ a finite point and an ideal point are distinct, and $\overline{\text{conv}}$ of $\{\kappa_*, \eta\}$ contains the Euclidean ray from κ_* in direction η ; its interior points $\kappa_* + r\eta$ ($r > 0$) lie on $\{z = 1\}$, so are in X only if they are contacts in K . Were the whole ray in K , the medial centres a_k — equidistant from q_k and q'_k with both feet ascending — would, by the monotonicity of position distance, have a foot at the projection of α_k to this convex contact ray, contradicting two separated feet for large k (no-collapse, Theorem 4.3). So some ray point is a witness, $\mathcal{H}$ is extended-defective. Finally $p \in \mathcal{H}$ by the drift (margin $\geq \delta$, Theorem 3.6). $\square$

Proposition 12.13 (Two escaping feet in $\mathbb{H}^2$). *Let $X \subseteq \mathbb{H}^2$ be closed and graph-type, $\mathcal{H} = \{z > 1\}$ the maximal free horoball of a drift reconstructing $p \notin \hat{X}$, and suppose both of the drift's feet escape ($|\beta_k|, |\beta'_k| \rightarrow \infty$). Then the feet run off to the two opposite ideal points, $K_\infty = \{-\infty, +\infty\}$, the extended hull is all of $\mathbb{R}$, and $\mathcal{H}$ is extended-defective and contains p : necessity holds.*

Proof. Both feet ascend by Lemma 4.2, so each escape direction $\eta = \lim \beta_k / |\beta_k| \in \{-\infty, +\infty\}$ lies in K_∞ . They are distinct: if $\beta_k, \beta'_k \rightarrow +\infty$ together, then for large k both feet lie on the single right-hand graph branch of X , on which (Theorem 4.3, F_t strictly convex) the nearest point of a centre is unique — contradicting two feet. So one foot runs to $-\infty$, the other to $+\infty$, and $K_\infty = \{-\infty, +\infty\}$. A closed convex subset of $\mathbb{R}$ whose recession set contains both $\pm\infty$ is all of $\mathbb{R}$, so $\overline{\text{conv}}(K \cup K_\infty) \cap \mathbb{R} = \mathbb{R} \neq K$ (a proper closed subset for any $X \neq S$): $\mathcal{H}$ is extended-defective. As p has depth $\delta^* > 0$, $p \in \mathcal{H} = \{z > 1\}$. $\square$

Remark 12.14 (Status of the residue). For compact X ($K_\infty = \emptyset$) the kernel is dissolved (§6.1, Theorem 6.3). The non-compact tail $K_\infty \neq \emptyset$ has sufficiency from [2, Thm. 4.1] (ambient defect), and necessity from Proposition 12.12 (one foot bounded, one escaping — the strand) and Proposition 12.13 (both feet escaping, $\mathbb{H}^2$), the ascent of escaping feet now being automatic (Lemma 4.2). Thus in $\mathbb{H}^2$ the unified Conjecture 12.11 is a theorem, both directions, compact and non-compact alike. For $n \geq 3$ the non-compact case was the last holdout, since the direction-only ideal-contact set cannot distinguish a single asymptotic ridge (no reconstruction) from two parallel ones (reconstruction) — Example 12.15. Refining the ideal contact by its perpendicular *offset* and reading the moving sites at finite height closes it too (Theorem 6.10, §6.2): the unified statement is now a theorem for every graph-type set in every dimension. (The compact case is closed in every dimension, §6.1.)

12.3 The offset, and the boundary-envelope analysis

The direction-only ideal-contact set cannot distinguish a single asymptotic ridge from two parallel ones. The example below isolates the missing datum — the perpendicular offset — and the development that follows led, after one false turn (a conformal-sphere envelope) and one correction (the moving Euclidean diagram), to the picture of Part I.

For *anchored* non-compact sets the deepest-shadow method extends cleanly: when the deepest infimum is attained on a bounded set of minimisers (an attained contact realising the depth, with the bulk uniformly below the horosphere outside a compact set), the tilt of Lemma 6.1 applies verbatim — the attained minimisers rise at rate $\geq \varepsilon$ under an infinitesimal tilt toward the side on which π separates, contradicting maximality — so π lies in the recession-cone hull $\overline{\text{conv}}(K^*) + \text{cone}(K_\infty)$, and screening (Lemma 6.2, dimension- and compactness-free) finishes necessity as before. This covers the strand, the rotationally symmetric valley (whose rim supplies a full sphere $K_\infty = S^{n-2}$, numerically confirmed), and every set with an attained anchor.

But the direction-only data $K_\infty \subseteq \partial_\infty S$ is not enough when there is no anchor, and this is a genuine $n \geq 3$ obstruction, not a technicality. The witness is a pair of configurations in $\mathbb{H}^3$ that share *every* directional invariant yet behave oppositely.

Example 12.15 (Single versus parallel ridge). In $\mathbb{H}^3$ let

$$X_1 = \{(s, 0, 1 - \text{sech } s)\}, \quad X_2 = \{(s, c, 1 - \text{sech } s)\} \cup \{(s, -c, 1 - \text{sech } s)\} \quad (c > 0),$$

a single ridge and two parallel ridges, all running to the ideal direction $\eta = (1, 0)$ at height $\rightarrow 1$. Both have $K^* = \emptyset$ and the *same* ideal-contact set $K_\infty = \{\eta\}$. Yet $\mathcal{R}_X(X_1)$ reconstructs nothing near the horoball (a single graph branch is Chebyshev; numerically no point with two feet at separated parameters), while X_2 reconstructs the strip between its ridges (medial centres on $\{y = 0\}$ with one foot per ridge; numerically $(s, 0, z) \in \mathcal{R}_X$ for $z > 1$). Any invariant built from K^* and the *directions* K_∞ alone — in particular the recession-cone hull — assigns both the same value and so cannot be correct.

This retracts the closure claimed in an earlier draft: the recession-cone hull captures the right *shape* (a sector, a ray, a line) but not its *location*. We now formulate the correct object and prove the mechanism that resolves Example 12.15.

In Example 12.15, $\eta = (1, 0)$, $\eta^\perp$ is the y -axis; for X_1 , $B_\eta = \{0\}$ (convex — not ray-defective); for X_2 , $B_\eta = \{+c, -c\}$ (non-convex — ray-defective). So the offset set distinguishes them where the direction cannot.

Proposition 12.16 (Offset symmetry forces reconstruction). *Let $X \subseteq \mathbb{H}^n$ be invariant under the reflection of $\eta^\perp$ fixing a hyperplane $H \ni 0$, and suppose $B_\eta \not\subseteq H$ (two asymptotic rays swapped by the reflection, with offsets off H). Then the fixed locus $\{(\alpha, t) : \alpha^\perp \in H\}$ consists of medial points (reflection pairs a foot on one ray with its mirror on the other), and their balls reconstruct an open subset of $\{z > 1\}$: $\mathcal{R}_X \neq \emptyset$, witnessing ray-defectiveness. The single-ray X_1 , having no such symmetry and a Chebyshev (singleton) offset set, reconstructs nothing near the horoball.*

Proof. The reflection σ of $\eta^\perp$ extends to a hyperbolic isometry fixing ∞ and the locus $L = \{\alpha^\perp \in H\}$. For $a \in L$, σ fixes a and permutes X , so the nearest point and its mirror are equidistant from a ; as $B_\eta \not\subseteq H$ these are two *distinct* feet (on the two swapped rays), so a is medial. Letting a run up L gives medial balls whose union is a non-empty open set, exactly as for two ridges in distinct directions (§12.2, numerically the whole strip). For X_1 the foot map on a single asymptotic graph is single-valued, so no medial point arises near the horoball. $\square$

The remaining notes record the moving-diagram analysis as it was developed: the multi-direction picture obtained by completing the square, the closed-form sufficiency, the assessment of a proposed boundary-envelope unification, and the honest status at each stage. The conclusions are exactly those proved cleanly in §6.2; the discussion here documents the route and the dead ends, in particular why confinement of the drift is neither available nor needed.

This is the promised reduction, now a theorem: the multi-scale obstruction is real but inert, because the $\log t$ -scale penalty is common to all rays of one profile and cancels. Numerically the medial offsets track Motzkin exactly — for $B_\eta = \{-2, 0, 2\}$ they converge to the non-Chebyshev points ± 1 as the height grows, and the convex continuum $B_\eta = [-2, 2]$ reconstructs nothing.

The moving-contact picture (multi-direction). Completing the square in Lemma 6.5 unifies *all* the cases — several directions, unequal profiles, mixed attained/asymptotic — into a single Euclidean Motzkin problem with *moving* contacts. Indeed, with $\psi(\chi) \sim e^{-\lambda\chi}$ the minimising arclength is $\chi^* \sim \frac{2}{\lambda} \log t$, and the expansion collapses to

$$\cosh d(a, \ell_{\eta, \lambda, b}) = 1 + \frac{(t-1)^2}{2t} + \frac{|\alpha - P_{\eta, \lambda, b}(t)|^2}{2t} + o(1/t), \quad P_{\eta, \lambda, b}(t) := \frac{2}{\lambda}(\log t)\eta + b,$$

the same form as for an attained contact κ (which has $P \equiv \kappa$). *So at height t every contact — attained or asymptotic — presents as a point of $\mathbb{R}^{n-1}$:* the attained ones fixed at κ , the asymptotic ones *escaping* along their direction η at speed $\frac{2}{\lambda} \log t$, offset by b . The identity is exact in the limit (verified numerically: the ratio of the two sides tends to 1 to four places by $t = 10^3$). Consequently a high centre $a = (\alpha, t)$ has its feet at the Euclidean-nearest effective positions, and is medial iff α is non-Chebyshev for the moving set

$$K(t) := K^* \cup \{P_{\eta, \lambda, b}(t)\} \subseteq \mathbb{R}^{n-1}.$$

This is the correct form of the boundary-envelope idea: not a power diagram on $\partial_\infty \mathbb{H}^n = \mathbb{S}^{n-1}$ — on which every asymptotic ray collapses to the *single* pole ∞ (bounded height, position $\rightarrow \infty$), so directions and offsets are invisible — but an ordinary Voronoi/Motzkin diagram in $\mathbb{R}^{n-1}$ whose sites *move* as $t \rightarrow \infty$. The fragmented cases are now one statement: *a single direction* gives sites $\frac{2}{\lambda}(\log t)\eta + b$ on a common escaping line, so $K(t)$ is non-convex iff the offsets B_η are (Theorem 6.6); *many directions* give sites escaping along distinct rays, whose convex hull has a vertex per extreme direction; *unequal rates* make the smaller- λ sites escape faster, so they reach

the hull first and dominate (the slower cells vanish); and *attained contacts*, being fixed while the asymptotic sites recede to infinity, are overtaken unless they themselves sit on the hull. Gemini’s envelope intuition is thus right, with the conformal sphere replaced by the moving Euclidean diagram and the weight replaced by the escape speed $\frac{2}{\lambda} \log t$.

Sufficiency is the whole horoball, not a swept region. One might expect the reconstructed set to be a thin region swept by the moving medial centres — a horospherical cylinder over the offset bisector. It is not: numerically, parallel ridges at offset $\pm c$ reconstruct *all* of $\{z > 1\}$ (every offset, far ahead, behind), exactly as the circle does. The reason is the same high-centre mechanism, and it gives sufficiency in closed form.

So defect implies the whole horoball, exactly as in the compact case — no region refinement, against Point 3 of the boundary-envelope proposal. The proof used the confinement $|\alpha(t)| = O(\log t)$ of the *chosen* witness centres — the Motzkin non-Chebyshev point of $K(t)$, which we are free to take inside $\text{conv } K(t)$: this keeps *their* effective-position expansion valid, since beyond $|\alpha| = o(t)$ the cosh ceases to be quadratic. But confinement here is a property we *impose* on the witnesses we pick, not one that holds for all reconstructing centres, and the stronger reading of Point 1 — that *every* medial centre is so confined, provable by comparison against a fixed attained contact — is false. It is vacuous in the open case, where there is no attained contact to compare against ($K^* = \emptyset$, as for the parallel ridges of Example 12.15); and it fails outright, because the entire bisector plane of two asymptotic rays consists of medial centres, each reconstructing the whole horoball however far it has drifted laterally (numerically, centres at lateral distance up to $75 \log t$ from the sites still reconstruct a fixed p , with two feet on the two rays). Confinement is therefore not the route to necessity.

The route that does work is the no-collapse mechanism of Theorem 4.3, now applied to a single ray.

Remark 12.17 (Honest status). What is now *proved*: the offset — not just the direction — is the correct refinement (Definition 6.4, Example 12.15); and for asymptotic approach in a *single* direction with a common profile, defect reduces exactly to Motzkin in $\eta^\perp$ (Theorem 6.6), the medial offsets tracking the non-Chebyshev points (numerically exact). The supposed multi-scale obstruction turned out to be inert: the $\log t$ -scale ray penalty is common to all rays of one profile and cancels under the envelope theorem (Lemma 6.5). Moreover the whole cross-talk now has a clean form: by completing the square, an asymptotic ray presents at height t as a contact at the moving point $P_{\eta,\lambda,b}(t) = \frac{2}{\lambda}(\log t)\eta + b$ (identity verified to four places), so several directions, unequal profiles, and mixed attained/asymptotic configurations all become one Euclidean Motzkin problem on the moving set $K(t)$ — now *both* halves proved. Sufficiency: $K(t)$ non-convex implies the *whole* horoball is reconstructed (Theorem 6.7), with no swept-cylinder refinement. Necessity: a reconstructing drift’s two feet sit on two distinct sites at every height (Theorem 6.10), so $K(t)$ is non-convex — read at *finite* height, where the moving site of a ray is $P_r(t) = \chi^*(t)\eta + b(\chi^*(t))$. The earlier “non-merging” worry was a mis-statement of the gap: uniform separation of the limiting rays is *not* needed and can even fail, as converging ridges $\pm c/\chi$ show (Example 6.9) — their sites coalesce yet they reconstruct, because $K(t)$ is non-convex at each finite t . Confinement of the drift, contrary to the boundary-envelope proposal’s Point 1, is neither available (the bisector plane reconstructs at every lateral distance) nor needed. So the hyperbolic case is now closed for every graph-type set in every dimension — compact or not, anchored or not — both directions (Theorem 7.1). What lies strictly beyond is only the non-graph regime (wild, non-semiconvex asymptotic approach), where the decomposition into

convex-profile pieces fails, and the move off $\mathbb{H}^n$ (§13).

13 Toward non-hyperbolic Hadamard manifolds

The $\mathbb{H}^n$ theory rested on two pillars that fail in general, and naming their failure delimits the next step precisely.

Pillar 1: homogeneity. “WLOG $\mathcal{H} = \{z > 1\}$ ” used the transitive action of $\text{Isom}(\mathbb{H}^n)$ on horoballs. This survives only in two-point homogeneous (rank-one symmetric) spaces and Damek–Ricci spaces; for a general Hadamard manifold each horoball must be treated in place.

Pillar 2: flat horospheres. The whole Motzkin reduction — feet of high centres are the intrinsic-nearest contacts, and convexity of the contact set is Euclidean — used that horospheres of $\mathbb{H}^n$ are intrinsically flat. The Gauss equation locates the obstruction quantitatively: for a horosphere S with shape operator Π (positive semidefinite, as horoballs are convex), a tangent 2-plane has intrinsic curvature

$$K_S = K_M + \det(\Pi|_{\text{plane}}).$$

In $\mathbb{H}^n$, S is umbilic, $\Pi = \text{Id}$ and $K_M = -1$, so $K_S = -1 + 1 = 0$: flat, and Euclidean Motzkin applies. Under pinching $-b^2 \leq \sec \leq -a^2 < 0$ one has principal curvatures in $[a, b]$, hence

$$K_S \in [a^2 - b^2, b^2 - a^2],$$

which can be *positive*. So horospheres are in general not CAT(0), and the foundational equivalence “Chebyshev $\Leftrightarrow$ convex” (Theorem 4.1) need not hold — on positively curved spaces Chebyshev sets can be non-convex. This, not homogeneity, is the deep obstacle.

What should survive. The structural backbone is curvature-free or robust: completeness gives $\widehat{X} \setminus X \subseteq \mathcal{R}_X$ [2, Thm. 3.1]; rigidity (Proposition 3.3), the drift theorem (Theorem 3.6, via the horofunction boundary), and the structure trichotomy (Theorem 3.8) are stated for all Hadamard M . What must be rebuilt is the local dictionary at the horosphere:

- (i) *Feet $\rightarrow$ intrinsic-nearest contacts.* As a centre drifts to ξ along the axis through the horosphere (which it meets orthogonally), the comparison of $d_M(a, \cdot)$ on S_γ should localise to the induced metric d_{S_γ} ; this is the general replacement for the monotonicity used throughout §4 and §5 and appears provable from Jacobi-field control of the second fundamental form.
- (ii) *A Chebyshev/convexity theorem in S_γ .* With (i), both sufficiency and necessity need only: in the induced geometry of the horosphere, a closed contact set with a centre having ≥ 2 nearest points is non-convex, and conversely. This is Motzkin’s theorem in an Alexandrov space with two-sided curvature bounds — true under suitable hypotheses, the precise class being the question.

Two natural arenas present themselves, with different trade-offs; which to develop first is the fork below.

- *Rank-one symmetric spaces* (CH^n , HH^n , OH^2) and Damek–Ricci spaces: homogeneity (Pillar 1) is retained, horoballs are still a single orbit, but horospheres are nilpotent groups (Heisenberg-type) with left-invariant metrics — non-flat, sub-Riemannian in the limit. Motzkin becomes a Chebyshev question in a Carnot group.

- *Pinched negative curvature* (general $-b^2 \leq \sec \leq -a^2$): no homogeneity, horospheres Alexandrov with two-sided bounds; the cleanest setting for (i)–(ii) above as pure comparison geometry, at the cost of explicit models.

14 Summary and open problems

Result	Hypothesis	Status
$\widehat{X} \setminus X \subseteq \mathcal{R}_X$	complete	proved [2, Thm. 3.1]
defective $\mathcal{H} \subseteq \mathcal{R}_X$	Hadamard	proved [2, Thm. 4.1]
No escape from medial centres (rigidity)	Hadamard	proved (Prop. 3.3)
Drift $\Rightarrow$ maximal $\mathcal{H} \ni p$, contacts captured	Hadamard	proved (Thm. 3.6)
Trichotomy hull/drift/ray-escape, no residual case	Hadamard	proved (Thm. 3.8)
$\mathcal{R}_X \setminus \widehat{X} \subseteq \text{drift}$	$\mathbb{R}^n$	proved (Prop. 3.10)
Drift or asymptotic tangency	Hadamard	proved (Thm. 10.1)
Split upgrade, non-degenerate drift	$\mathbb{H}^2$	proved (Prop. 10.4)
Defective $\setminus$ split $\neq \emptyset$ (strand)	$\mathbb{H}^2$	example (Ex. 10.3)
Split sufficiency, $\eta > L_\infty/2$	$\mathbb{H}^2/\text{pinched}$	proved (Thm. 9.5, Cor. 9.6)
Split-or-defective conjecture	$n \geq 3$	false (Thm. 11.1)
Splitness is a 2-dim artefact	—	Rem. 11.2
Horospherical defect: circle test	$\mathbb{H}^3$	proved (Prop. 11.3a)
Horospherical defect = defective	$\mathbb{R}^n$	proved (Prop. 11.3b)
Horospherical defect $\approx$ split	$\mathbb{H}^2$	proved (Prop. 11.3c)
Corrected conjecture	Hadamard	open (Conj. 11.4)
Horosph. defective $\mathcal{H} \subseteq \mathcal{R}_X$, X on horosphere	$\mathbb{H}^n$	proved (Thm. 5.1)
X convex on horosphere $\Rightarrow \mathcal{R}_X = \emptyset$	$\mathbb{H}^n$	proved (Thm. 5.1)
Horosph. defective $\mathcal{H} \subseteq \mathcal{R}_X$, isolated contacts	$\mathbb{H}^n$	proved (Thm. 5.2)
Circle: $\mathcal{R}_X = \text{union of horosph. defective}$	$\mathbb{H}^3$	proved (Cor. 11.5)
Sufficiency, tangential bulk / curved horosph.	Hadamard	open (Rem. 12.8)
Contact feet $\Rightarrow$ defect (Motzkin)	$\mathbb{H}^n$	proved (Prop. 12.2)
Ascent of bounded feet to K	$\mathbb{H}^n$	proved (Lem. 12.3)
Non-merging estimate $\Rightarrow$ necessity	$\mathbb{H}^n$	proved (Thm. 12.4)
No collapse at any graph contact	$\mathbb{H}^n$	proved (Thm. 4.3)
Deepest-shadow lemma	$\mathbb{H}^n$ compact	proved (Lem. 6.1)
Necessity, compact graph-type X	$\mathbb{H}^n$	proved (Thm. 6.3)
Corrected conj., compact graph-type X	$\mathbb{H}^n$	theorem (Cor. 7.2)
Equidistant-flow kernel	$\mathbb{H}^n$ compact	dissolved (§6.1)
Unified conj. (extended hull) on strand/circle	$\mathbb{H}^n$	verified (Conj. 12.11)
Necessity, non-compact, one escaping foot	$\mathbb{H}^n$	proved (Prop. 12.12)
Necessity, two escaping feet	$\mathbb{H}^2$	proved (Prop. 12.13)
Ascent of all feet (bounded or escaping)	$\mathbb{H}^n$	proved (Lem. 4.2)
Unified conj., all of $\mathbb{H}^2$	$\mathbb{H}^2$	theorem (Rem. 12.14)
Non-compact, anchored deepest inf	$\mathbb{H}^n$	proved (§6.2)
Direction-only hull fails (single vs. parallel ridge)	$\mathbb{H}^3$	example (Ex. 12.15)
Offset set B_η distinguishes them; symmetry recon.	$\mathbb{H}^n$	proved (Def. 6.4, Prop. 12.16)
Single direction, common profile = Motzkin in $\eta^\perp$	$\mathbb{H}^n$	proved (Thm. 6.6)
Moving-contact identity $P = \frac{2}{\lambda}(\log t)\eta + b$	$\mathbb{H}^n$	proved (§6.2)
Defect $\Rightarrow$ whole horoball (sufficiency)	$\mathbb{H}^n, n \geq 3$	proved (Thm. 6.7)
No collapse on a single ray	$\mathbb{H}^n$	proved (Prop. 6.8)
Converging ridges: finite-height site, not limit offset	$\mathbb{H}^3$	example (Ex. 6.9)
Necessity: two feet = two distinct moving sites	$\mathbb{H}^n, n \geq 3$	proved (Thm. 6.10)
Moving-contact defect, both directions	$\mathbb{H}^n$ graph-type	theorem (Thm. 7.1)

Question 14.1 (The necessity gap beyond $\mathbb{H}^n$). In $\mathbb{H}^n$ this is now settled for graph-type sets (Theorem 6.10, Theorem 7.1); the questions below concern the move to a general Hadamard manifold, where the moving-Motzkin diagram is unavailable. Theorem 11.1 retires the split-or-defective form of the question; the live version is whether Conjecture 11.4 holds: for $p \in \mathcal{R}_X \setminus \widehat{X}$, must some maximal free horoball containing p be horospherically defective (Definition 2.3)? By Proposition 3.3 the upgrade cannot come from a single inflation: the medial witness of p injects its further contacts through equidistant motion of centres, as in the pincer of Theorem 9.5. The concrete target is an equidistant-flow theorem: starting from $a \in M_X$ and $p \in B(a, d(a))$, grow free balls with centres on the equidistant locus of the contact set, and show the family drifts to a horoball whose horospherical contact hull contains the shadow of p . For $n = 2$ this is the split upgrade of §10 (Remark 10.5 delimits its open core); for $n \geq 3$ the single segment J_t of the planar pincer must become a swept $(n - 1)$ -dimensional wall over $\text{hconv}_\gamma(K_\gamma)$.

Question 14.2. Formulate the correct asymptotic notion of horospherical contact hull — the flat convex hull in $S_\gamma \cong \mathbb{E}^{n-1}$ of the asymptotic contact set $\overline{\{x \in X : b_\gamma(x) \rightarrow c_\gamma\}}$, in a horospherical compactification — so that Conjecture 11.4 both covers the asymptotic strand (Example 10.3, Proposition 8.4(b)), where contacts recede to the ideal boundary, and reduces to Definition 2.3 when contacts are attained. Then identify, for a horospherically defective horoball, the exact reconstructed subregion (the second caveat after Conjecture 11.4), and prove the resulting equality.

Question 14.3. Theorem 3.4 uses only global minimality of the extended foot-geodesic. On a complete, non-Hadamard manifold the statement survives whenever the inflation ray is globally minimizing; the flat cylinder (where bitangency without multiplicity is possible, $X = S^1 \times \{0\}$) shows that bitangency must then be read as “two minimizing geodesics”. Determine the sharp generality.

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